BIOMECHANICS (EPHE 341)

Linear Kinematics

Describing objects in linear motion

Readings: McGinnis Chapter 2 pg 51–67

Learning objectives for this section

- Distinguish between linear, angular and general motion
- Define distance traveled and displacement
- Define average, instantaneous speed, velocity and acceleration
- Understand the value of linear kinematics in analyzing human movement.

Kinematics

Kinematics:

> The study of motion exclusive of the influences of mass and force. It includes displacement, velocity, and acceleration without regard for the forces acting on a body.
>
> the branch of biomechanics that deals with the description of the **spatial** and **temporal** components of motion

Three important Questions:

1. How far did an object/person travel?
2. How fast did the object/person travel?
3. Was the object/person moving faster at some points of the movement and slower at others?

In Biomechanics it is also important to consider the movement of the individuals limb segments

Human Kinematics

Human Kinematics

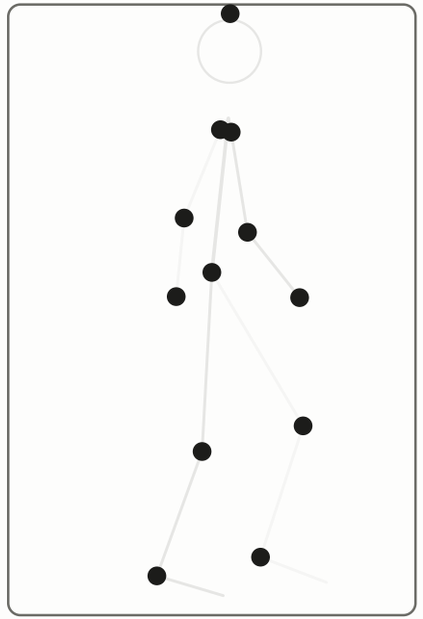
The study of Human Movement involves detailed kinematic measurement.

Important to understand normal human movement as well as following injury or during extreme performance.

Kinematic measures are different based on gender, affect, culture… what else?

http://www.biomotionlab.ca/Demos/BMLwalker.html

In order to understand human movement kinematics we first need to go back to the basics of linear kinematics



Source: one frame of a measured walking stride — SUSU youth motion dataset · UVic · CC BY 4.0

Kinematics

Types of Motion

1. Linear Motion

- uniform motion of all parts of an object in the same direction and at the same speed
- more commonly known as ***translation***
- does not necessarily mean motion in a straight line:
  straight line → rectilinear
  curved line → curvilinear



Rectilinear Translation



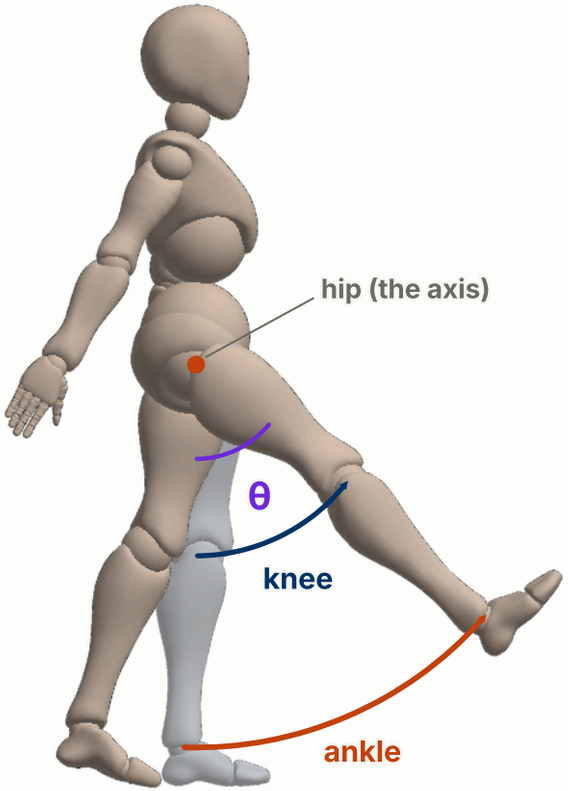
Curvilinear Translation

Kinematics

Types of Motion

2. Angular Motion

- rotation around an axis (axis can be real or imaginary)
- different parts of the object undergo different ***linear*** displacements
- all parts of the object undergo the same ***angular*** displacement
- for example, the ankle travels at a greater curvilinear speed than the knee
- both the ankle and the knee travel at the same angular speed



Kinematics

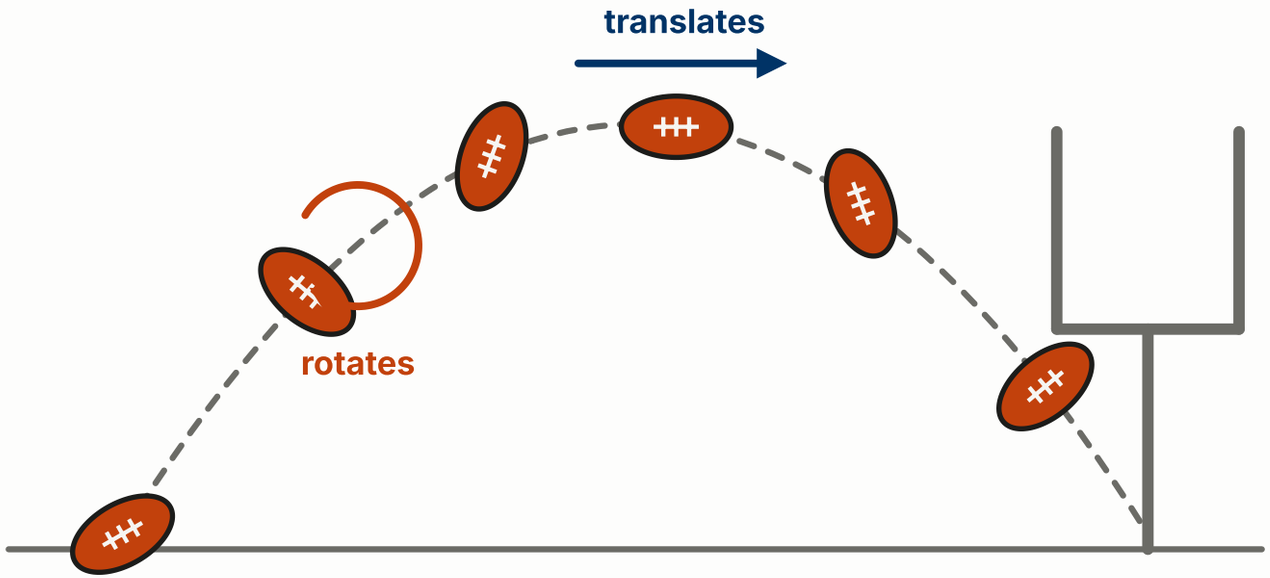
Types of Motion

3. General Motion

- a combination of linear and angular motion

Example:

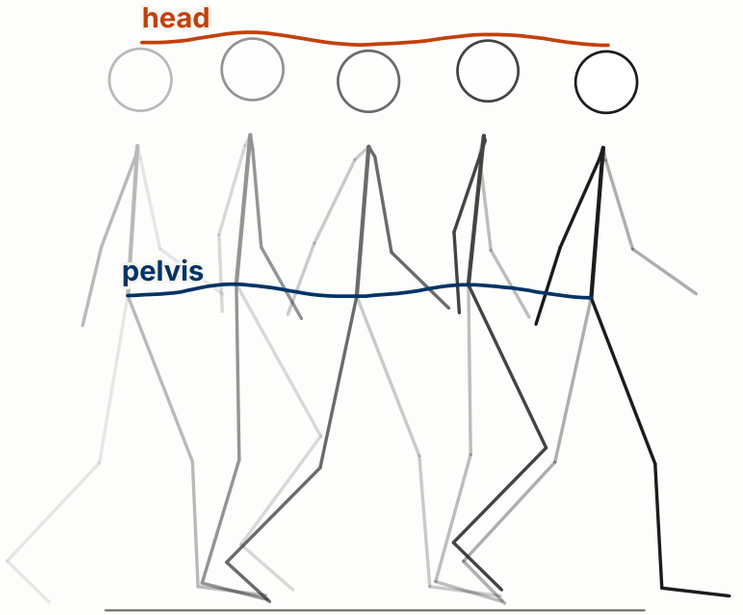
When kicked for a field goal, a football will translate towards the goal posts ***and*** rotate end-over-end.



Kinematics

Types of Motion · 3. General Motion

- Running and walking are good examples of general motion
- Angular motions of our limbs produce linear motions of our centre of mass and head
- In order to measure motion we need to consider **space** and **time**



Source: walking stride, subject SUSU-30 — SUSU youth motion dataset · UVic · CC BY 4.0

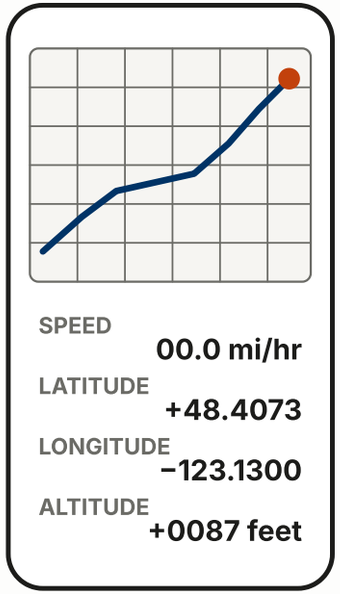
Spatial and Temporal

Position

- the most basic parameter in kinematics
- measured as the location within a fixed coordinate system
- in 2D, measured against an X- and Y-axis
- in 3D, X-, Y-, and Z- axes

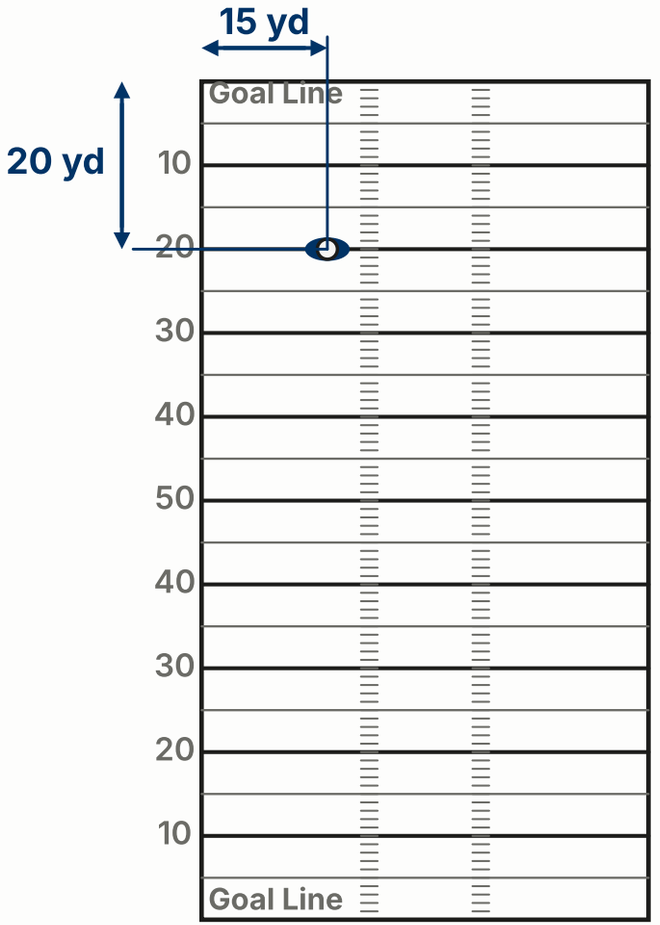
Quantifying location is space is difficult

This can be done using video, motion capture GPS, RFID



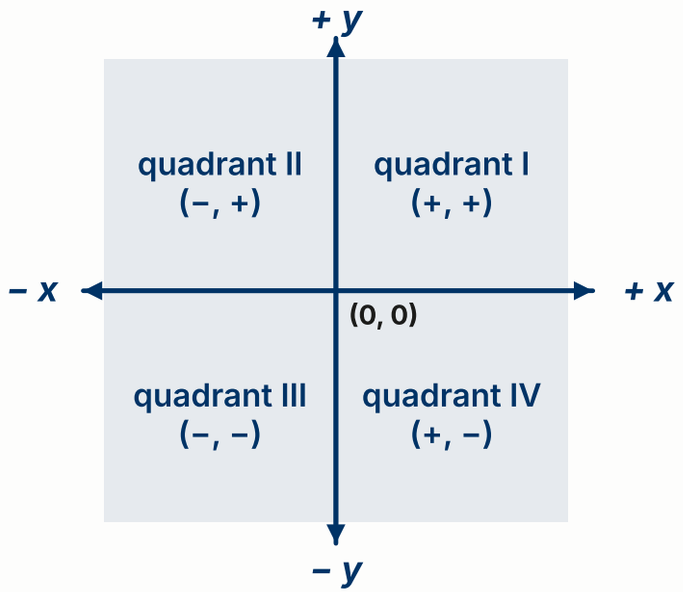
Position

How do you describe the Position of an object in space ?



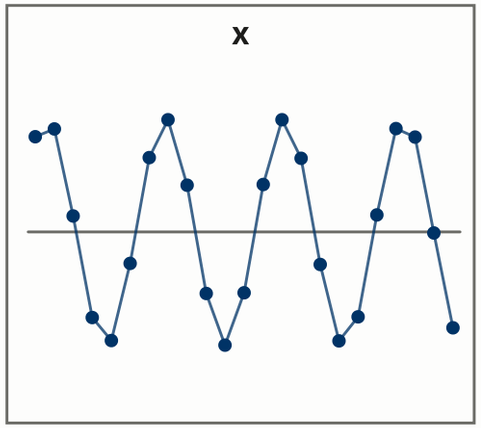
Cartesian Coordinate System

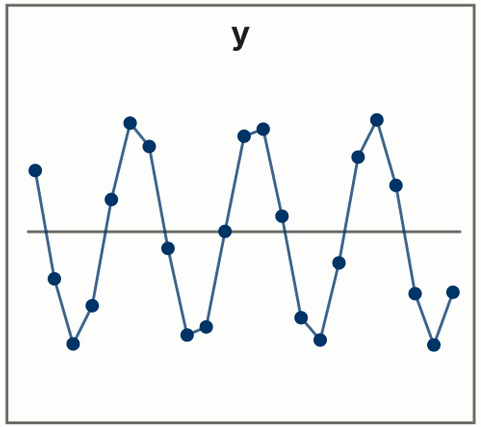
- an axis system used to define the position of a vector in 2D or 3D space is the Cartesian coordinate system (Ax, Ay) and the sign depends on the position relative to the axis origin (0,0)
- The origin is defined by a specific landmark

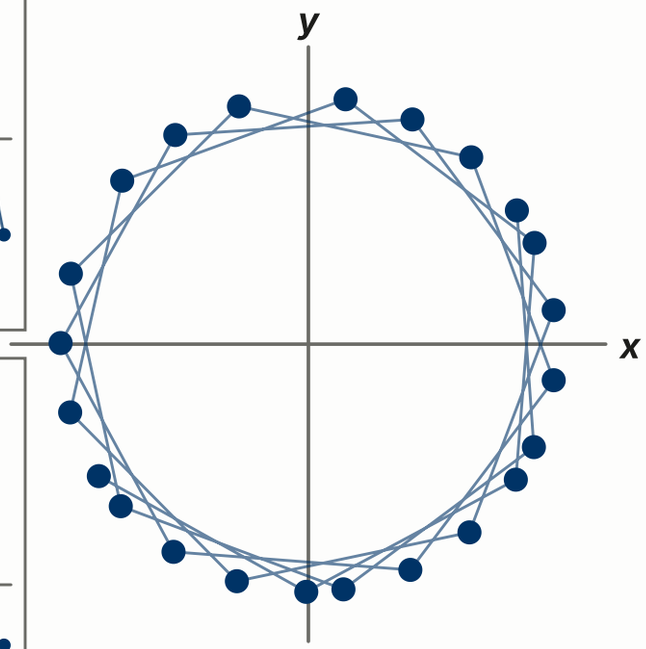


Cartesian Coordinate System

time	x	y
1	0.8415	0.5403
2	0.9093	-0.4161
3	0.1411	-0.9900
4	-0.7568	-0.6536
5	-0.9589	0.2837
6	-0.2794	0.9602
7	0.6570	0.7539
8	0.9894	-0.1455
9	0.4121	-0.9111
10	-0.5440	-0.8391
11	-1.0000	0.0044
12	-0.5366	0.8439
13	0.4202	0.9074
14	0.9906	0.1367
15	0.6503	-0.7597
16	-0.2879	-0.9577
17	-0.9614	-0.2752
18	-0.7510	0.6603
19	0.1499	0.9887
20	0.9129	0.4081
21	0.8367	-0.5477
22	-0.0089	-1.0000
23	-0.8462	-0.5328

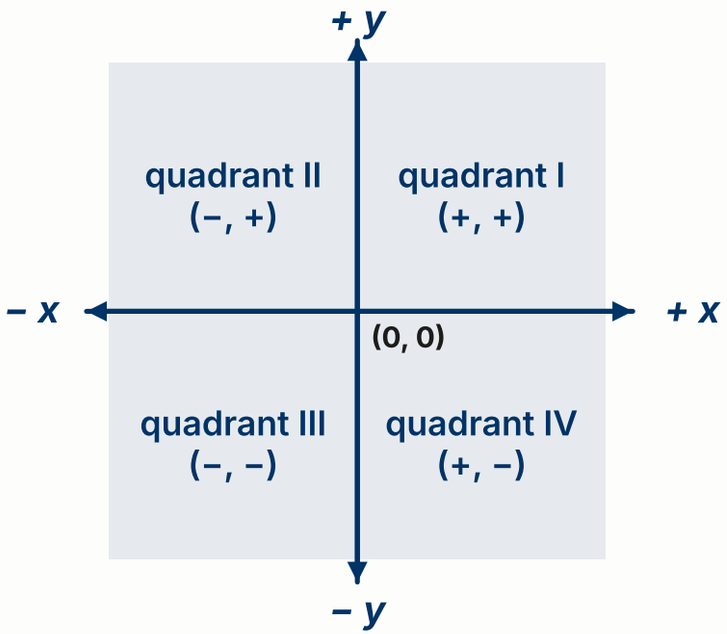


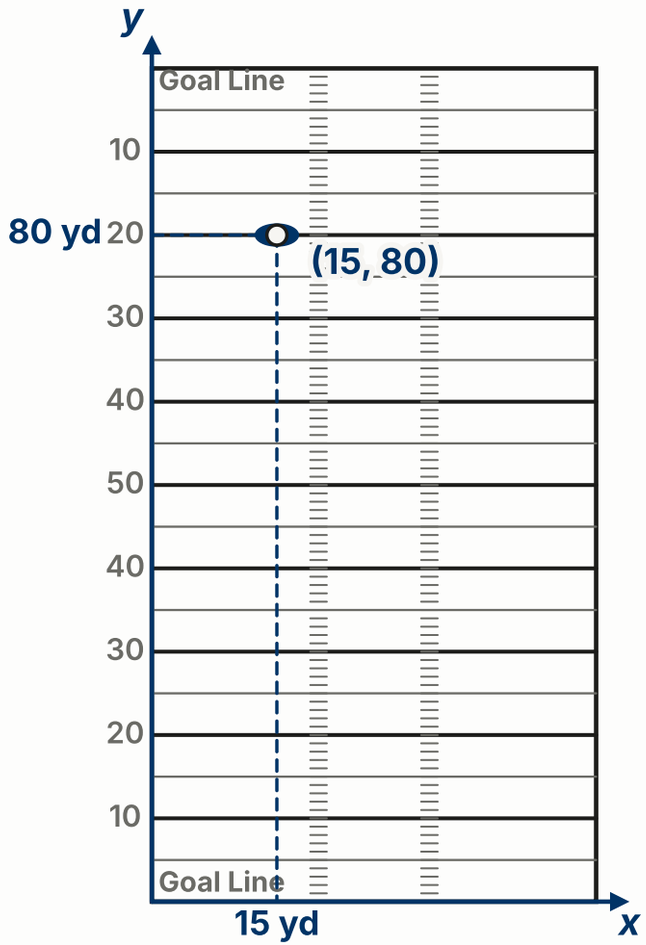




Cartesian Coordinate System

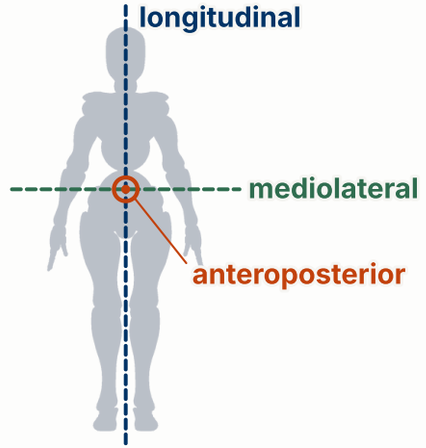
– Where is the origin in this example?



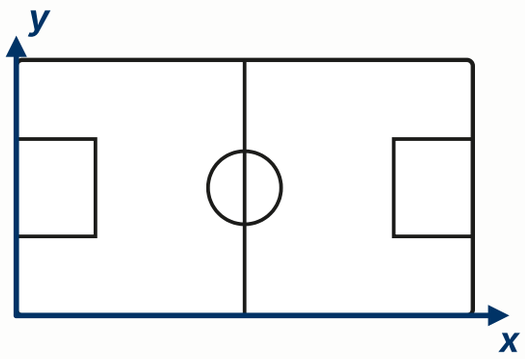


Cartesian Coordinate System

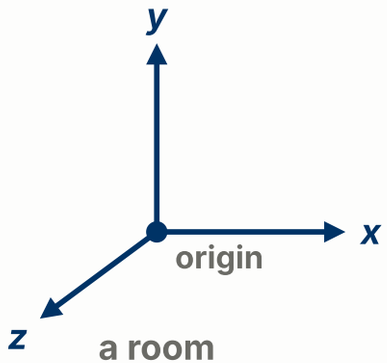
- An axis system is normally defined as perpendicular (orthogonal) reference axis such as
  - Horizontal and Vertical
  - Based on landmarks
    - Room, court, body segment
  - Orientation of the body or body segment
    - Anatomical planes and axis



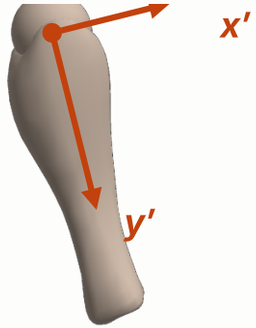
Source: character rig: Mixamo (Adobe)



a court



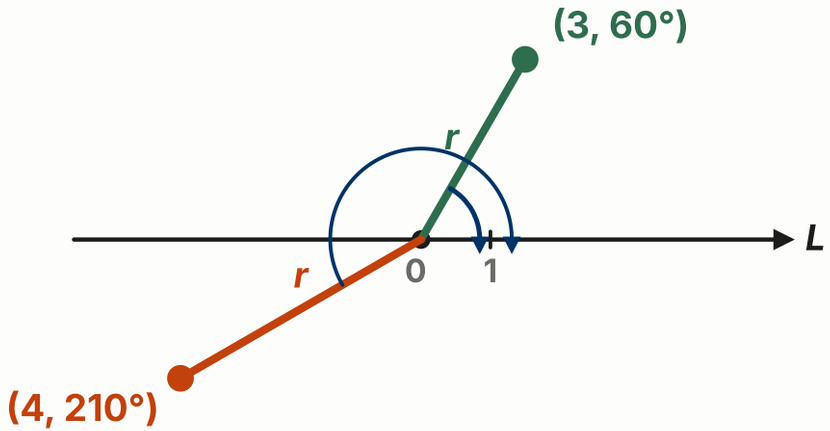
a room



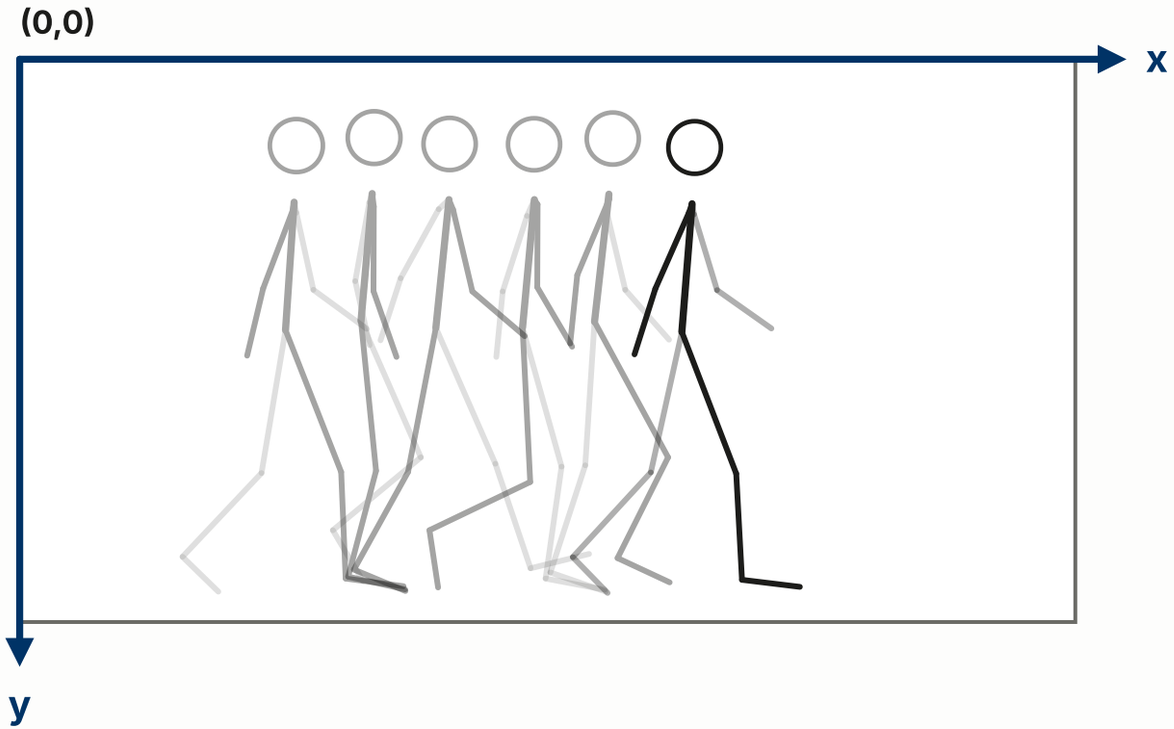
a body segment

Other coordinate Systems

- Polar Coordinate System
  - two-dimensional coordinate system in which each point on a plane is determined by a distance from a reference point and an angle from a reference direction.



Position - space

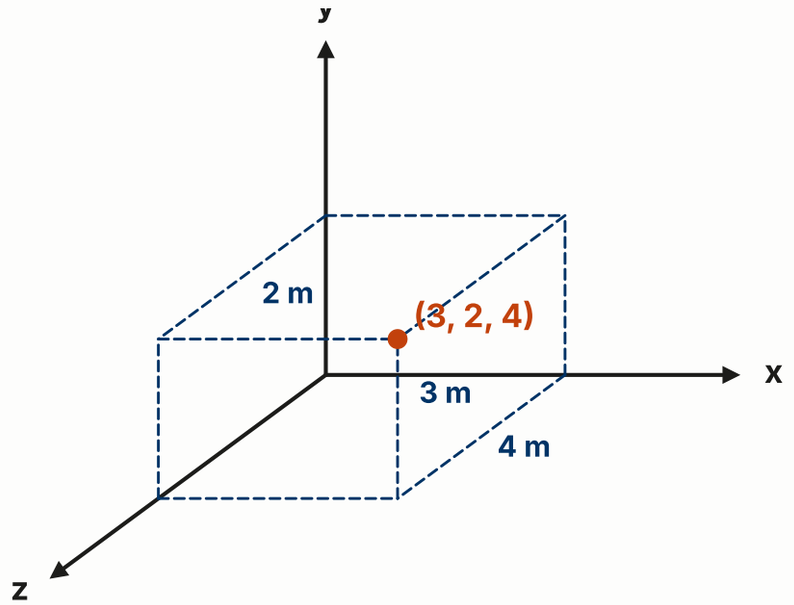


Source: walking stride, subject SUSU-30 — SUSU youth motion dataset · UVic · CC BY 4.0

- In image analysis the coordinate system normally relates to the visual field of an image and involves a calibration of image pixels to actual units of measurement. (pixels to the right, pixels down)

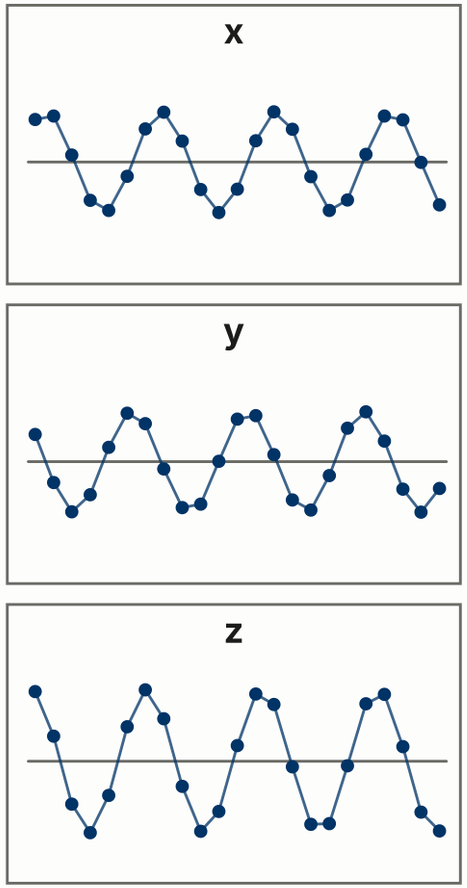
Position - space

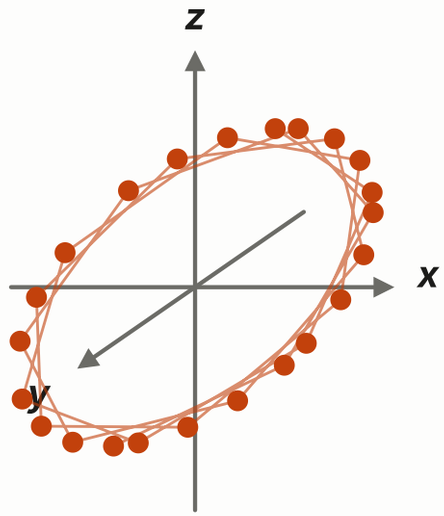
In 3 dimensional space you need 3 axis (x,y,z) and state the position of an object using 3 coordinates (x,y,z)



Position - space

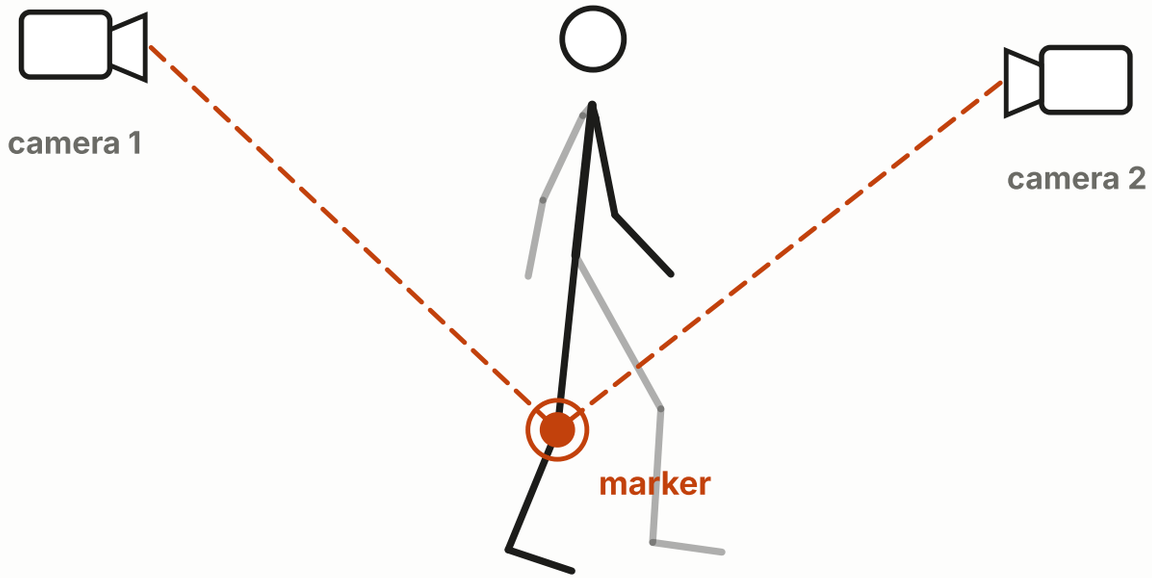
time	x	y	z
1	0.841	0.540	1.382
2	0.909	-0.416	0.493
3	0.141	-0.990	-0.849
4	-0.757	-0.654	-1.410
5	-0.959	0.284	-0.675
6	-0.279	0.960	0.681
7	0.657	0.754	1.411
8	0.989	-0.146	0.844
9	0.412	-0.911	-0.499
10	-0.544	-0.839	-1.383
11	-1.000	0.004	-0.996
12	-0.537	0.844	0.307
13	0.420	0.907	1.328
14	0.991	0.137	1.127
15	0.650	-0.760	-0.109
16	-0.288	-0.958	-1.246
17	-0.961	-0.275	-1.237
18	-0.751	0.660	-0.091
19	0.150	0.989	1.139
20	0.913	0.408	1.321
21	0.837	-0.548	0.289
22	-0.009	-1.000	-1.009
23	-0.846	-0.533	-1.379





Position - space

- In 3 dimensional optical motion capture, reflective markers are placed on a human body and the position of a marker in space is determined by the triangulation of the position of the marker relative to at least two cameras.



Source: walking stride, subject SUSU-30 — SUSU youth motion dataset · UVic · CC BY 4.0

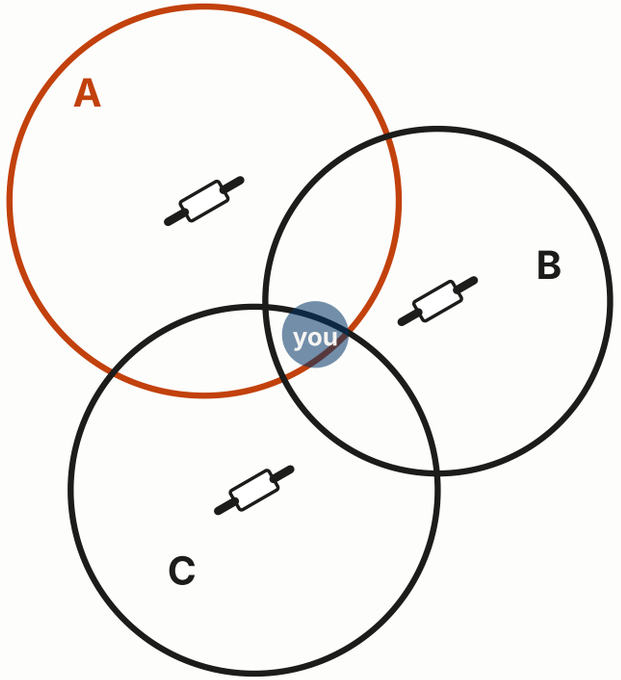
Position - space

- What about GPS?
  - Where we are on earth?

The Global Positioning System (GPS) is a network of about 30 satellites orbiting the Earth at an altitude of 20,000 km.

If you know how far away you are from satellite A, then you know you must be located somewhere on the red circle. If you do the same for satellites B and C, you can work out your location by seeing where the three circles intersect. This is just what your GPS receiver does, although it uses overlapping spheres rather than circles.

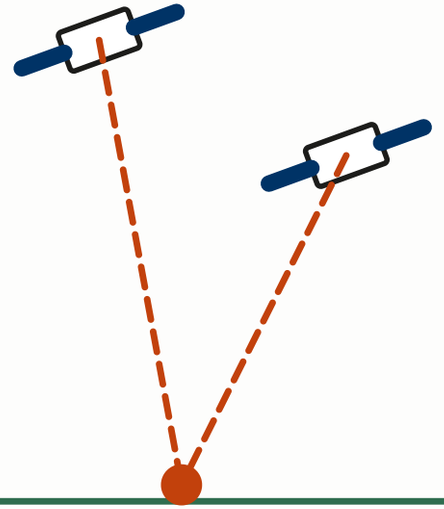
The more satellites there are above the horizon the more accurately your GPS unit can determine where you are.



Position - space

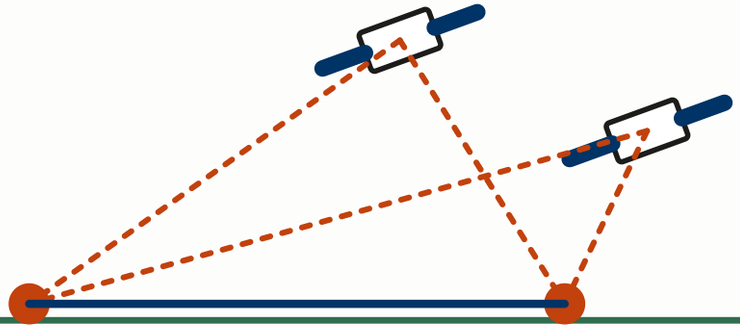
STANDARD GPS

Uses binary code signals
from multiple satellites
to determine position.



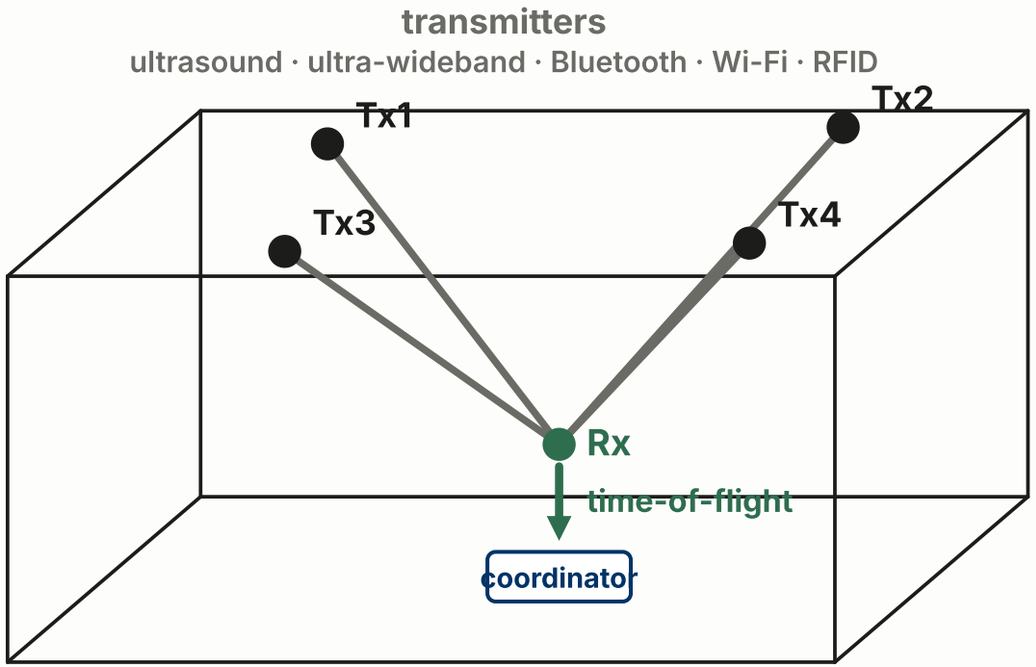
HIGH-PRECISION GPS

Uses multiple frequencies to
correct for atmospheric
effects on radio waves.
Uses carrier wave cycles
to compute precise vectors
between multiple receivers.



Position - space

- Indoor Location Sensing



Distance and Displacement

Change in position

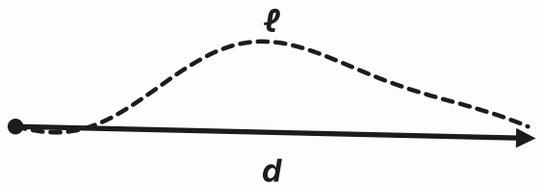
Linear Distance

- represented by the letter ℓ
- measured along the path of motion
- scalar variable (magnitude only)
- SI units: meters (m)
- no simple equation

Linear Displacement

- represented by the letter d
- measured in a straight line from the initial position to the final position
- vector variable (magnitude and direction)
- SI units: meters (m)

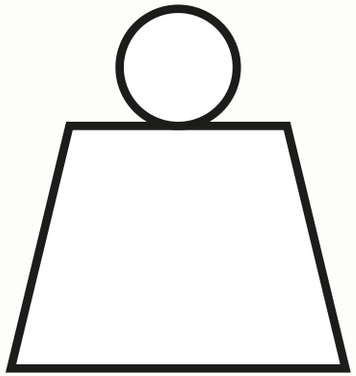
$$d = x_f - x_i$$
$$= \text{final position} - \text{initial position}$$



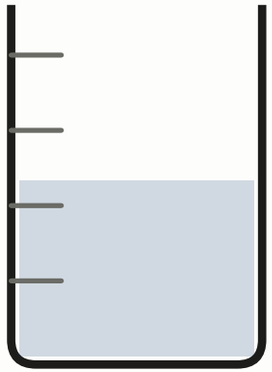
What are scalars and vectors?

Scalar

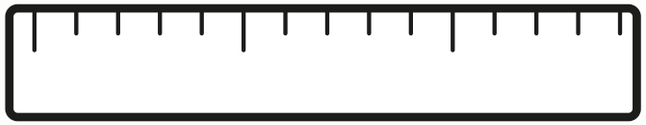
- a physical quantity that has a ***magnitude***, but no ***direction***
- eg: mass, volume, length



mass



volume



length

What are scalars and vectors?

Vector

- a physical quantity that has both a ***magnitude*** and ***direction***
- ***A simple arrow connecting two points***



- eg: displacement, velocity, acceleration, force
- in written form, typically designated by a letter with an arrow over it, although a half-arrow, or simply bold-faced type are also used

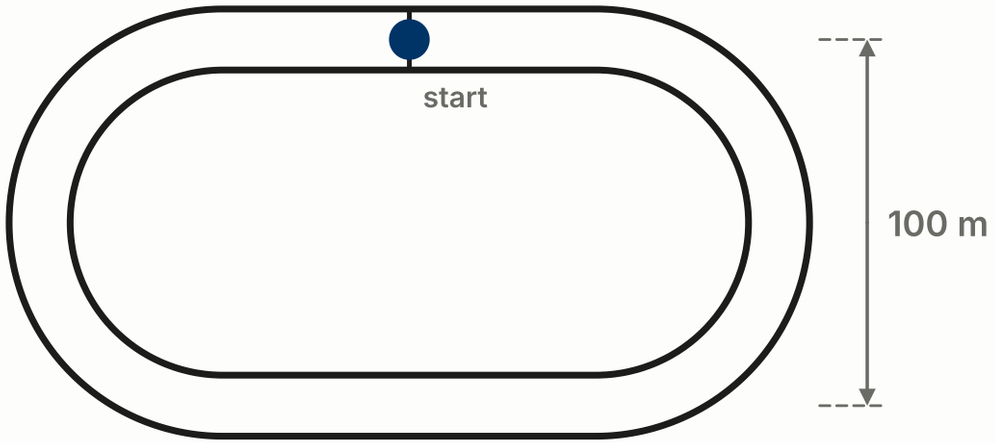
  $\vec{A}$ $\overrightarrow{A}$ $\mathbf{A}$

- the magnitude of a vector designated by a letter bound by vertical lines (absolute value), signifies that the magnitude of a vector is always positive $|\vec{A}|$

Distance and Displacement

Example Problem

A runner completes 1.5 laps of a 400 m track. If the track is 100 m wide, what are her distance and displacement?

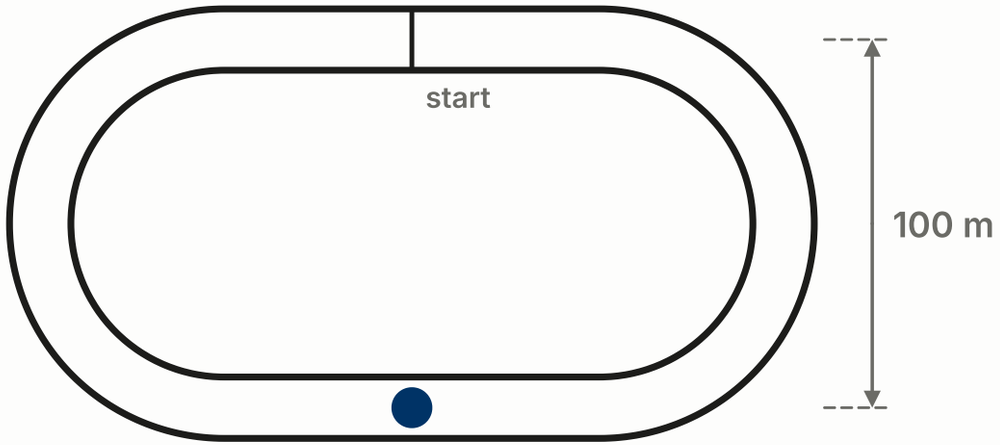


Distance and Displacement

Solution

Distance is the change in position, measured along the path of motion. Therefore, her distance travelled is:

$$400\ \text{m} \times 1.5\ \text{laps} = \mathbf{600\ m}$$

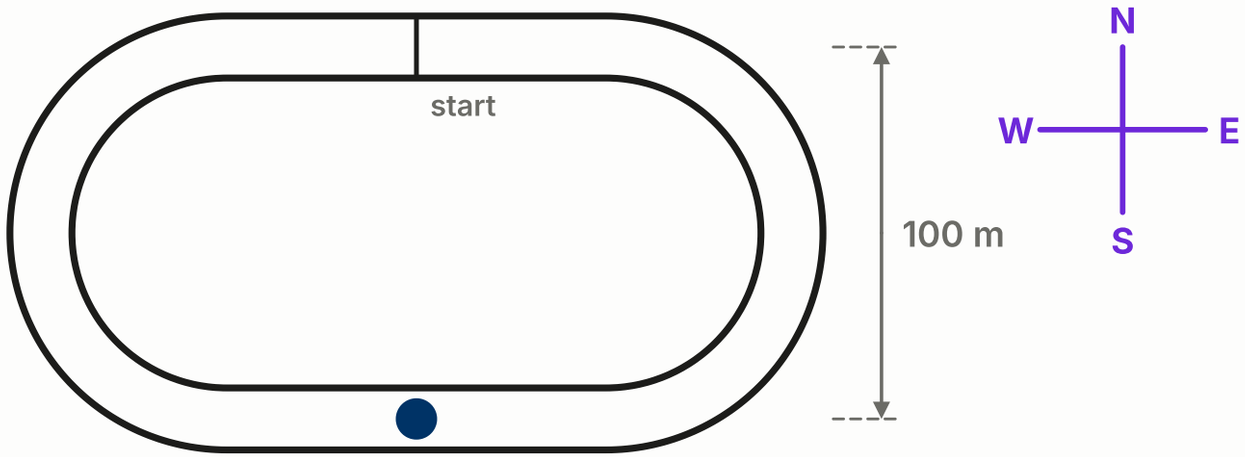


Distance and Displacement

Solution

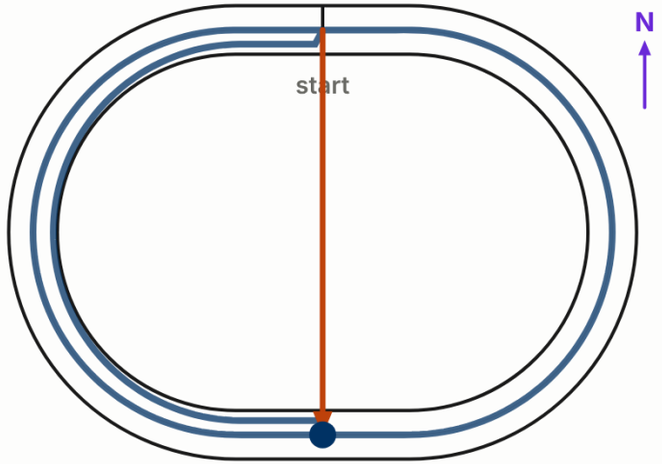
Displacement is the change in position, measured from the initial position to the final position. Therefore, her displacement is:

$$0\ \text{m} - 100\ \text{m} = \mathbf{-100\ m}\ \ (\text{or } 100\ \text{m south})$$



Distance and Displacement

The same runner, lap by lap



1.50 laps

distance run
600 m

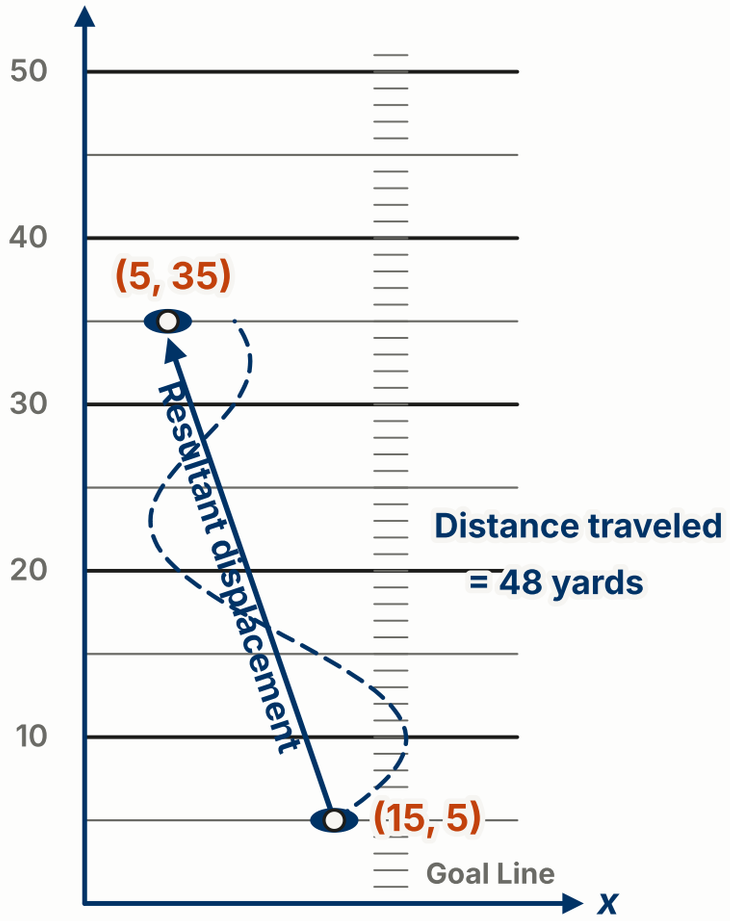
displacement
100.0 m south

Distance run — **600 m**

▶ Run

The **distance** is the whole path she has run; the **displacement** is the straight arrow from the start. At 1.5 laps: 600 m, and 100 m south.

Distance and Displacement



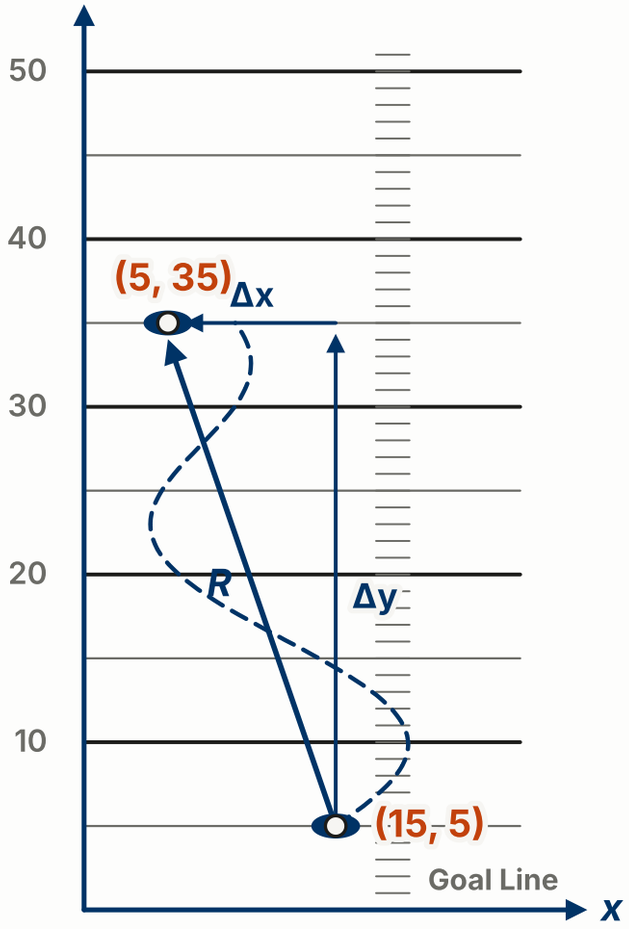
The previous example was simple because we were dealing with straightforward calculations.

What happens when we have coordinate locations?

Displacement is a vector which means it is the shortest (straight line) distance between the starting and ending position.

How do you determine the magnitude and direction of the resultant displacement vector in the following example?

Distance and Displacement



vectors can be easily determined by subtracting initial from final coordinates for respective Cartesian axes (x,y)

$$\text{Resultant } x = x_{\text{final}} - x_{\text{initial}}$$
$$= 5 - 15$$
$$x = -10$$

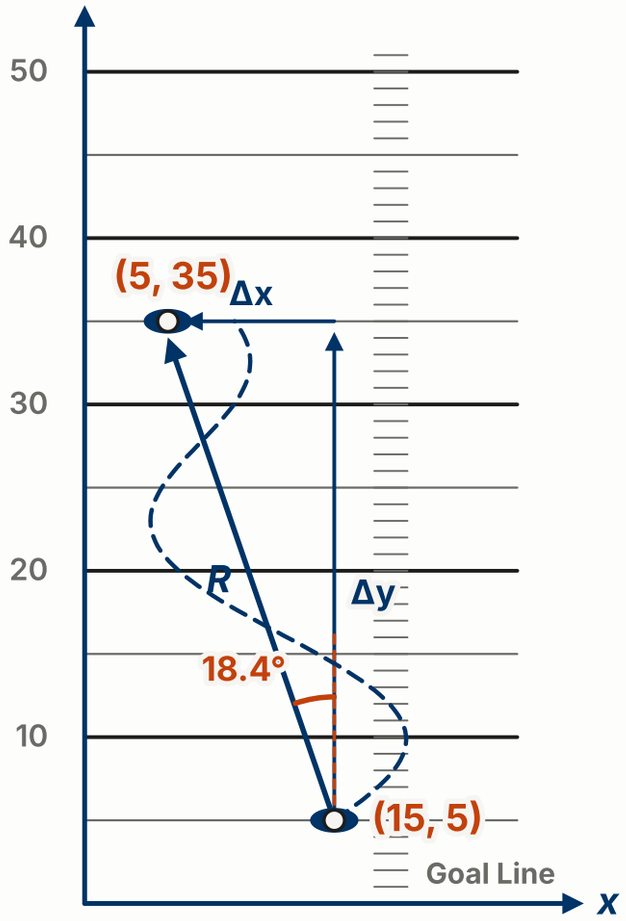
$$\text{Resultant } y = y_{\text{final}} - y_{\text{initial}}$$
$$= 35 - 5$$
$$y = 30$$

We can also use trig to find out the Resultant Vector Magnitude and Direction

Using the Pythagorean Theorem

$$A^2 = b^2 + c^2$$
$$A^2 = (-10)^2 + (30)^2$$
$$\mathbf{A = 31.6 \text{ yards}}$$

Distance and Displacement



Vector Direction

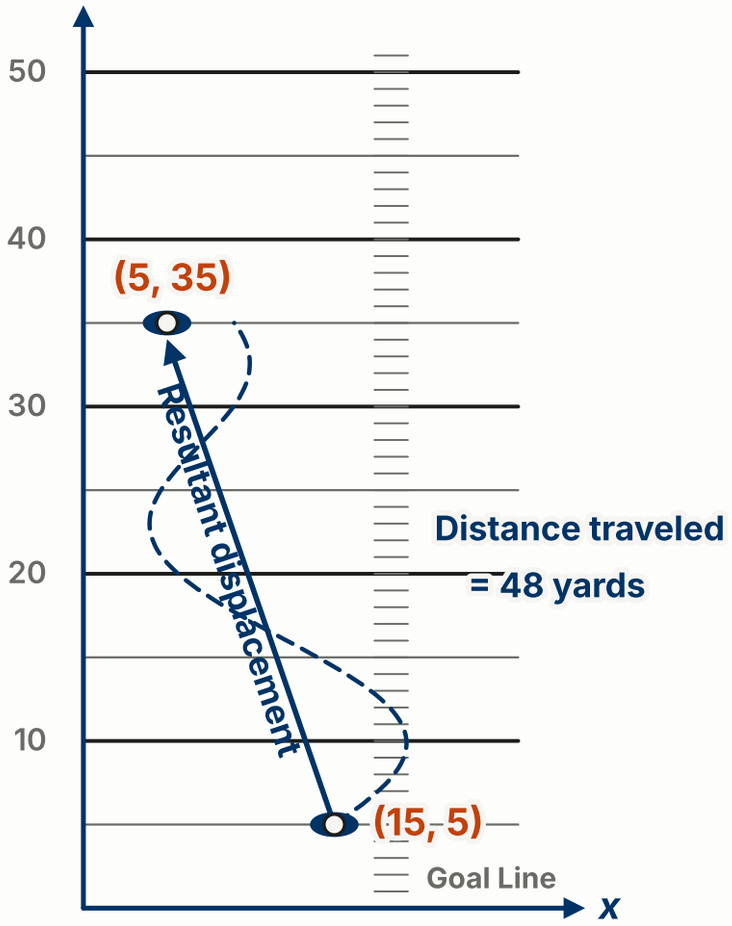
$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\tan\theta = \frac{-10}{30}$$

$$\theta = \arctan\left(\frac{-10}{30}\right)$$

$$\boldsymbol{\theta = -18.4 \text{ degrees}}$$

Distance and Displacement



When dealing with devices that will give us position coordinates the output can come in a few ways;

The data will be represented in rows and columns with samples and coordinates

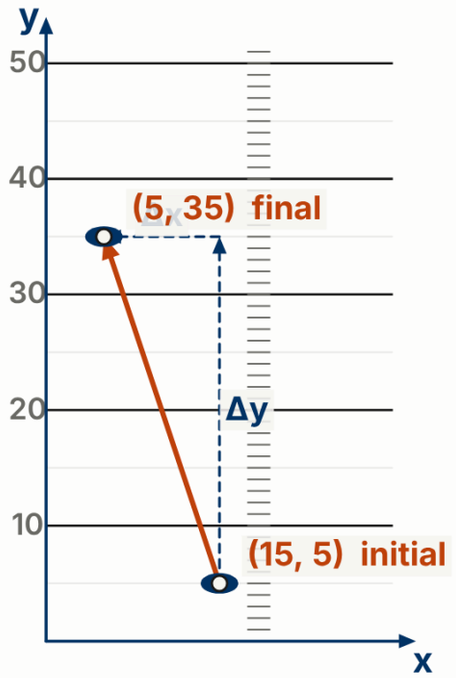
For example

	Column 1	Column 2
	(x-data)	(y-data)
Row 1 (time 1)	15	5
Row 2 (time 2)	5	35
Row 3 (time 3)	...	...

This will help us when using software to perform calculations for us.

Distance and Displacement

Any two positions



$$\text{Resultant } x = x_{\text{final}} - x_{\text{initial}} = 5 - 15 = -10$$

$$\text{Resultant } y = y_{\text{final}} - y_{\text{initial}} = 35 - 5 = 30$$

$$A^2 = (-10)^2 + (30)^2$$

$$\mathbf{A = 31.6 \text{ yards}}$$

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{-10}{30}$$

$$\boldsymbol{\theta = \arctan\left(\frac{-10}{30}\right) = -18.4^\circ}$$

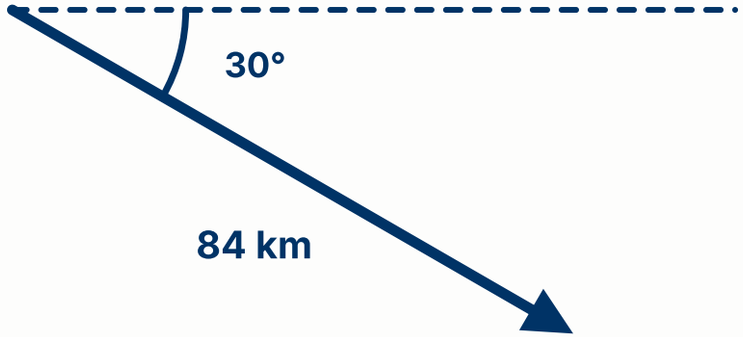
θ is measured from the +y axis; a negative angle leans towards −x

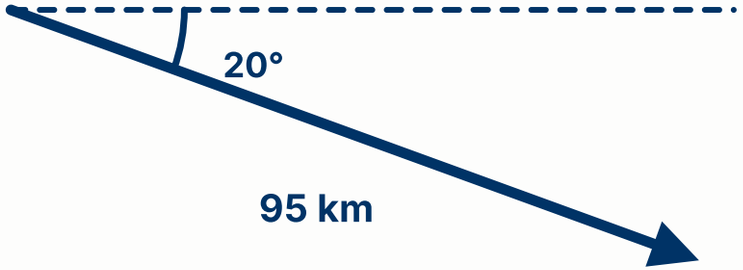
Back to the example

Drag either player. The displacement is **final minus initial** on each axis; its length comes from Pythagoras and its direction from the arctangent. The example: 31.6 yards at −18.4°.

Displacement and Vectors

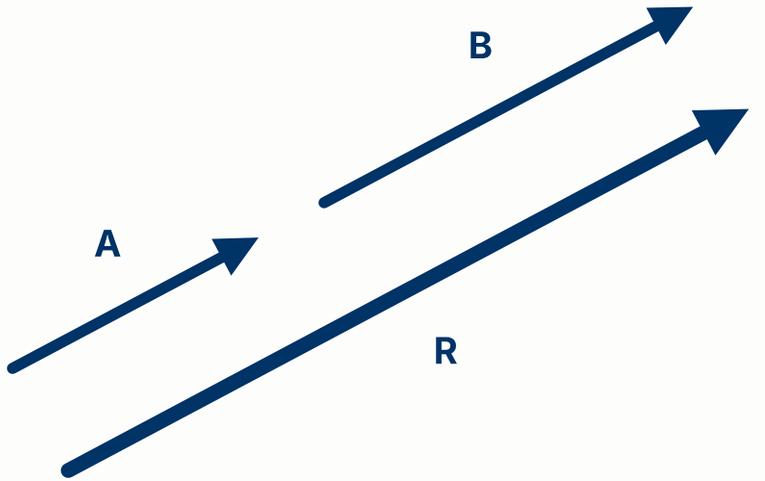
- The previous was a simple example using 1 vector
  - but what if we are dealing with more than one vector and need to add vectors.
  - What if we need to determine the x and y components of the displacement vector?





Vectors

- First we will deal with the simplest situation of collinear vectors
- two vectors are ***collinear*** if they *are parallel*
- The two vectors, **A** and **B** are collinear and may be added to form a ***resultant***
- collinear vectors can be added using simple arithmetic, or scalar addition
- the direction of the resultant vector is the same as that of vectors **A** and **B**



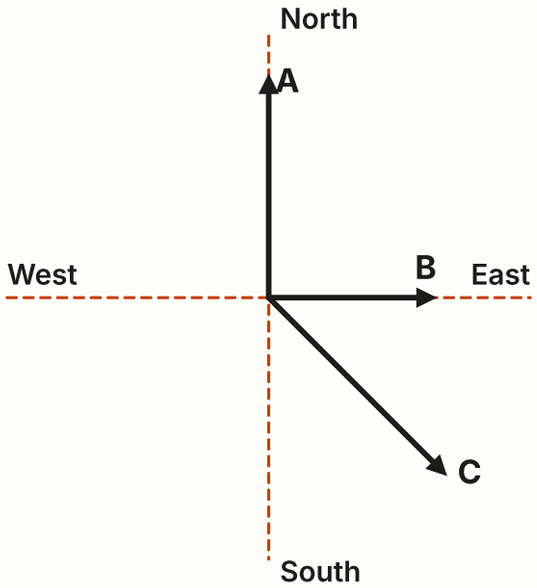
$$|\vec{A}| + |\vec{B}| = |\vec{R}|$$

More examples on page 27 of McGinnis in the Colinear Forces section

Vectors

- Non-collinear vectors
  - Are vectors that do not have points on the same line
  - Example; directions
- the addition of two or more non-collinear vectors can be accomplished using two methods:

1) Tip-to-Tail Method
2) Vector Resolution Method



Vectors

Tip-to-Tail Method

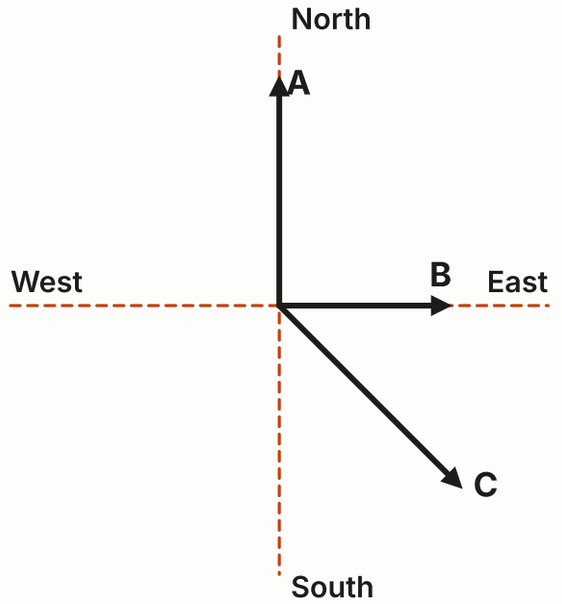
- [ ] Define your axis system and scaling factor.
- [ ] Draw each vector to scale with its tail on the tip of the preceding vector.
- [ ] Draw the resultant with its tail on the tail of the first vector and its tip on the tip of the last vector.
- [ ] Measure the length and orientation of the resultant vector to determine its magnitude and direction.

Vectors

Tip-to-Tail Method

for example

You are told to follow the directions in order of A, B, C. A is 4 km north, B is 3 km east and C is 4.5 km 45° south of east.



Vectors

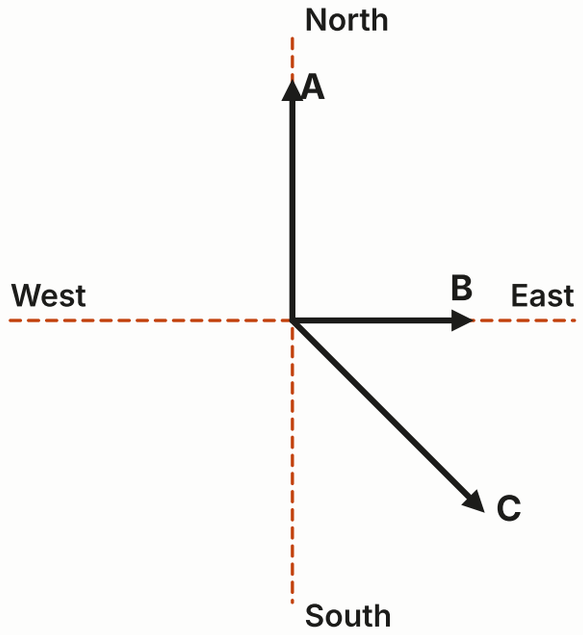
Tip-to-Tail Method

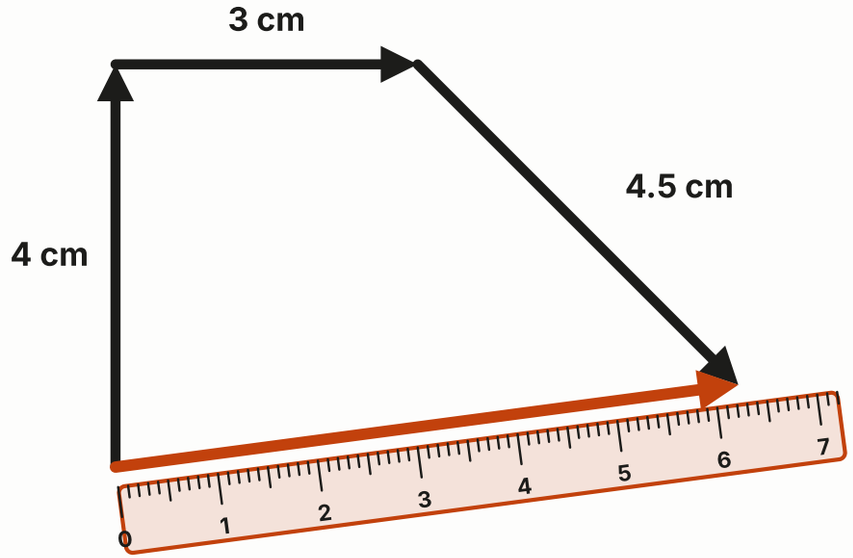
- Choose a scale that will allow you to fit the vectors onto your page
- For this problem, we will use a scale of 1 km : 1 cm

Vectors

Tip-to-Tail Method

Measure the length and orientation of the resultant vector to determine its magnitude and direction.





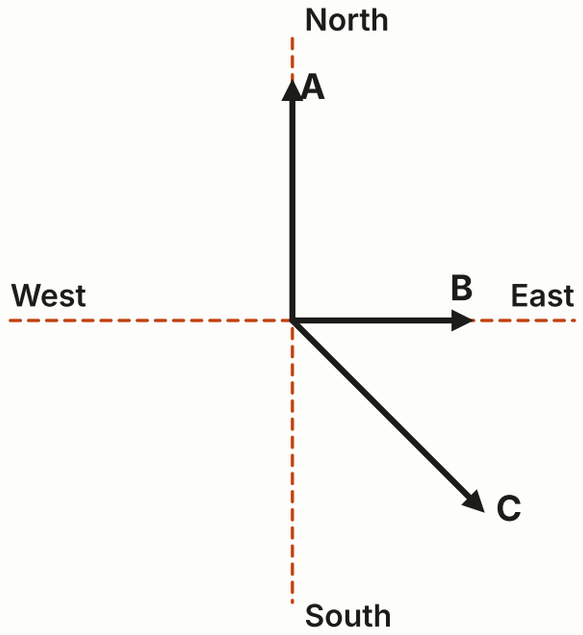
The resultant has a magnitude of

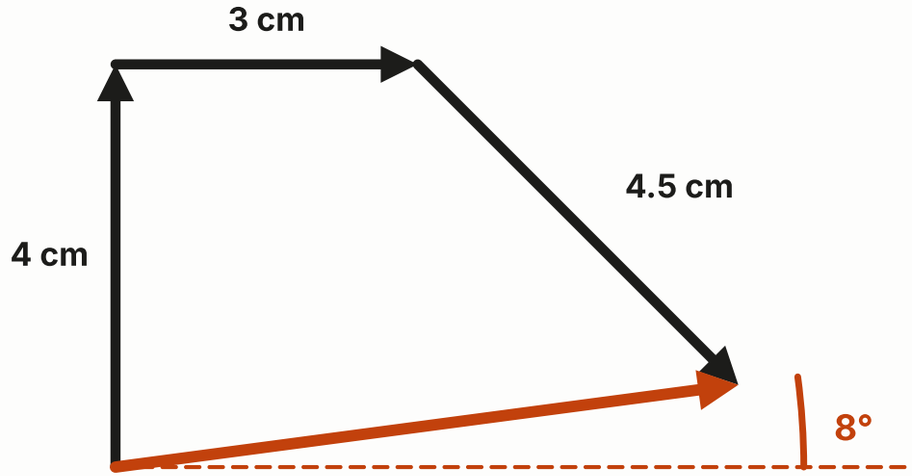
$$(\mathbf{6.2\ cm} \times \mathbf{1\ km}) = \mathbf{6.2\ km}$$

Vectors

Tip-to-Tail Method

Measure the length and orientation of the resultant vector to determine its magnitude and direction.

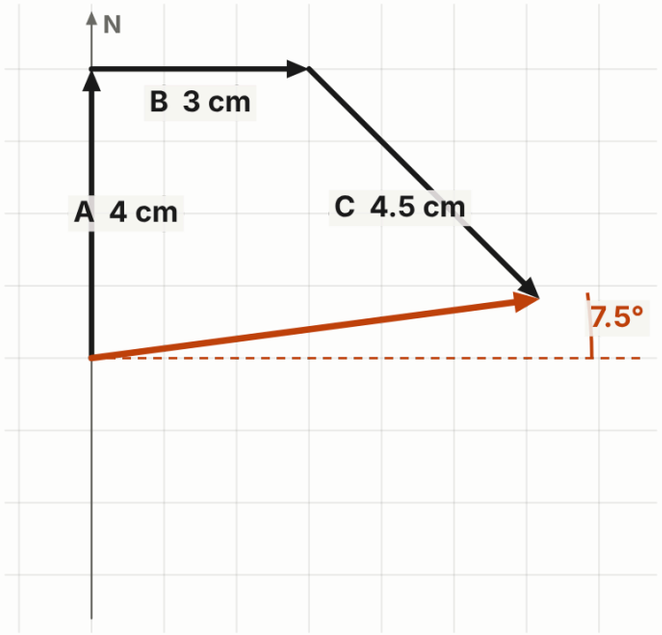




The resultant has a magnitude of **6.2 km** acting at an angle of **8° north of east**

Vectors

Tip-to-Tail Method



1. Define your axis system and scaling factor: 1 km : 1 cm
2. Draw each vector to scale with its tail on the tip of the preceding vector
3. Draw the resultant from the tail of the first to the tip of the last
4. Measure the length and orientation of the resultant

resultant 6.2 cm = 6.2 km

7.5° north of east

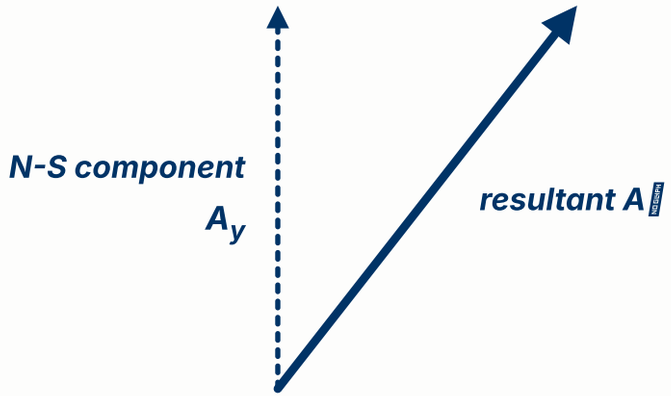
Step **4 of 4**

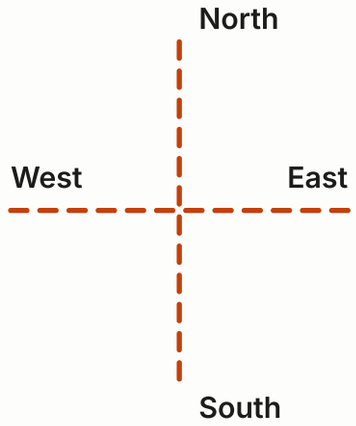
A, B, C	B, C, A	C, A, B	A, C, B

Drawn to scale, the resultant measures **6.2 cm**, so 6.2 km, at about **8° north of east**. Change the order: the path changes, the resultant does not.

Vectors

Vector Resolution Method

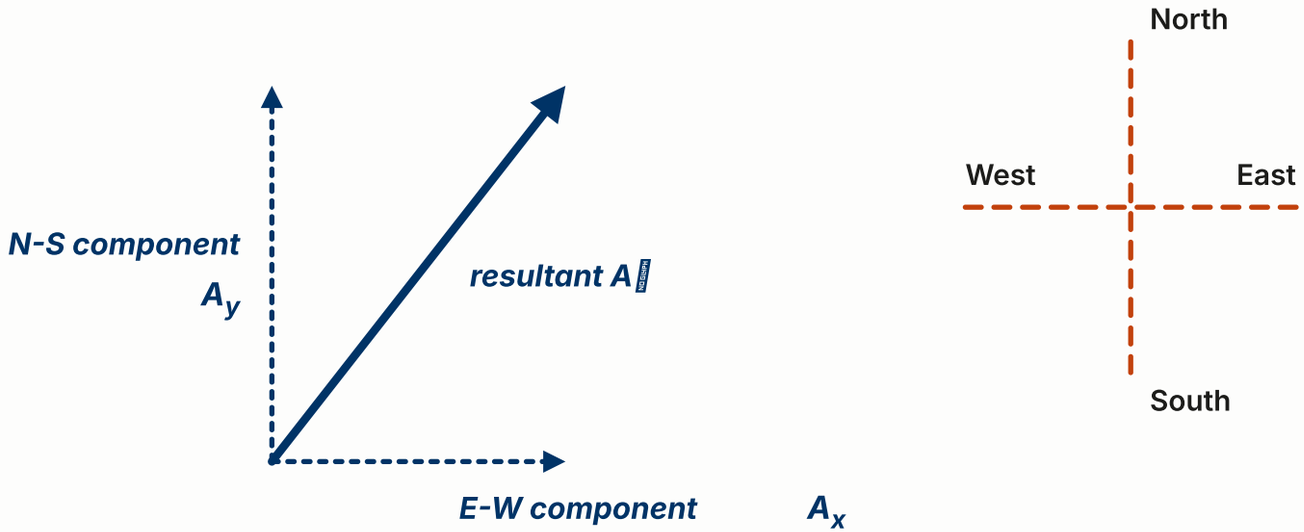




- vector resolution is the process of breaking down a vector into perpendicular components that can correspond to the coordinate system
- the North-South component of a vector can be represented graphically by drawing an arrow running N-S, beginning at the tail of the resultant and ending at the same N-S distance as the tip of the resultant

Vectors

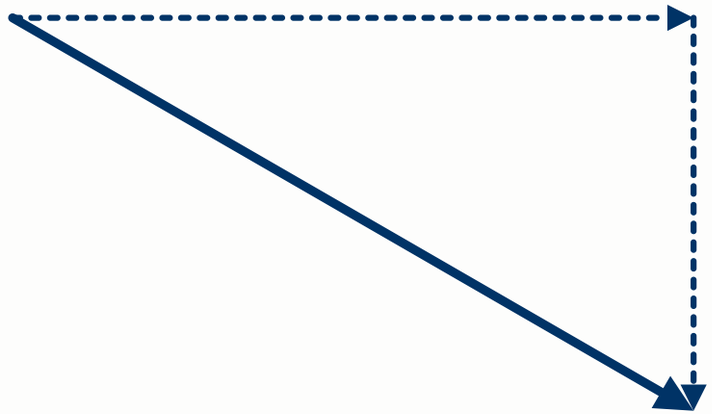
Vector Resolution Method



- The East-West component of a vector can be represented graphically by drawing an arrow running E-W, beginning at the tail of the resultant and ending at the same E-W distance as the tip of the resultant

Vectors

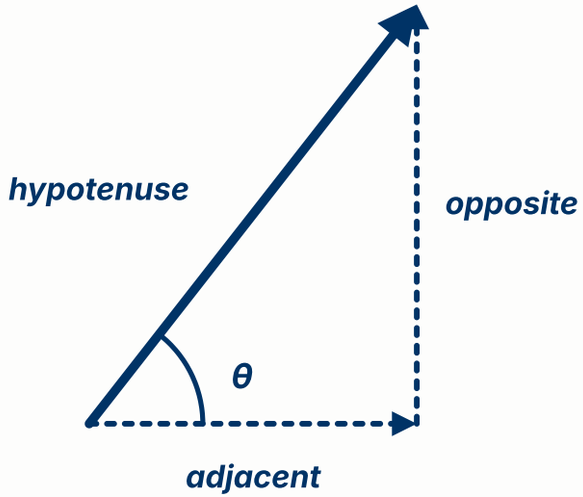
Vector Resolution Method



- The length of each component can be calculated using trigonometry
- Imagine moving one of the component vectors to form a right-angle triangle

Vectors

Vector Resolution Method

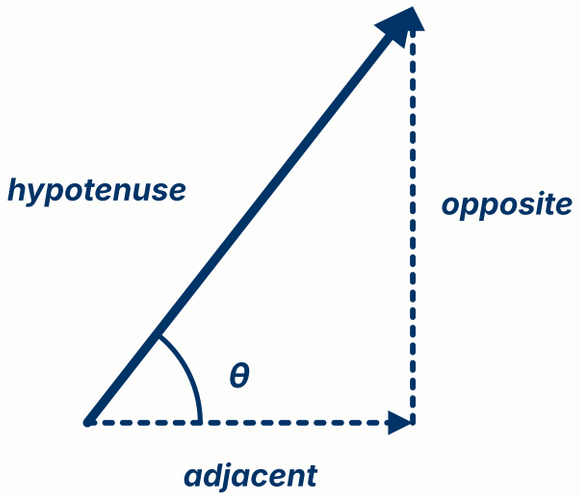


- if the angle, θ, is known, we can use the sine and cosine functions to determine the length of the components

$$\sin\theta = \frac{\text{opp}}{\text{hyp}} \qquad \cos\theta = \frac{\text{adj}}{\text{hyp}} \qquad \tan\theta = \frac{\text{opp}}{\text{adj}}$$

Vectors

Vector Resolution Method



- these equations can be rearranged to solve for the length (magnitude) of the vertical and horizontal components

$$\text{opp} = \text{hyp} \times \sin\theta$$

$$\text{adj} = \text{hyp} \times \cos\theta$$

$$\text{opp} = \text{adj} \times \tan\theta$$

Vectors

Vector Resolution Method

how to resolve a vector

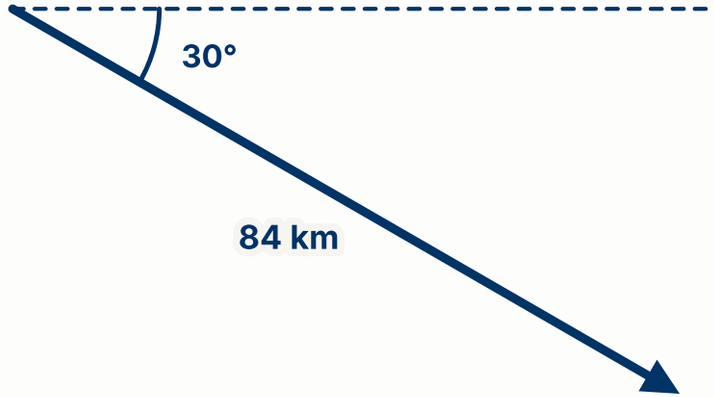
What are the N-S and E-W components of an 84 km displacement at an angle of 30° South of East?

Vectors

Vector Resolution Method

how to resolve a vector

1. Draw the vector being resolved

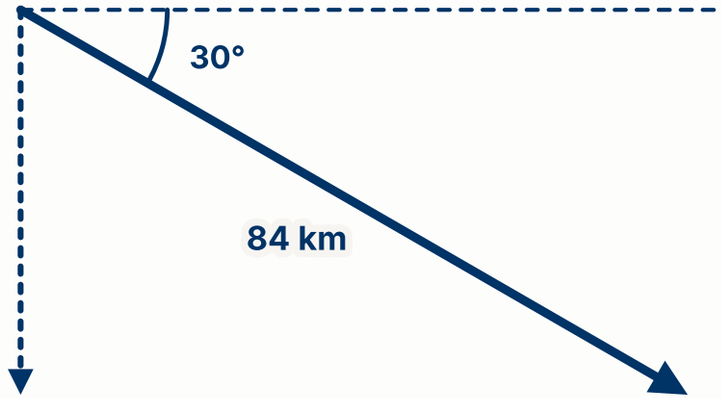


Vectors

Vector Resolution Method

how to resolve a vector

2. Draw the N-S component.

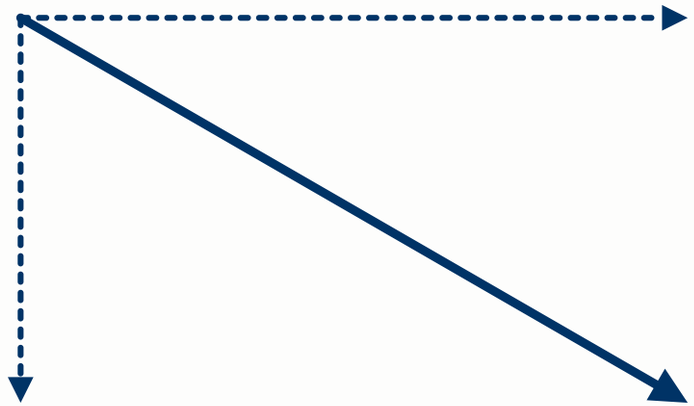


Vectors

Vector Resolution Method

how to resolve a vector

3. Draw the E-W component.

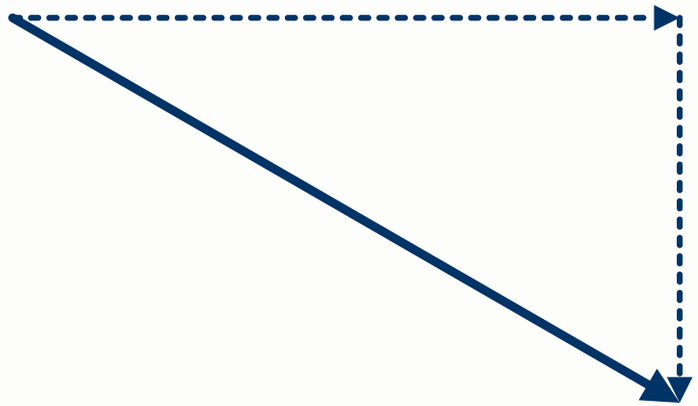


Vectors

Vector Resolution Method

how to resolve a vector

4. Imagine moving one of the component vectors to form a right-angle triangle.



Vectors

Vector Resolution Method

how to resolve a vector

5. Use the appropriate trigonometric functions to calculate the length of the components.

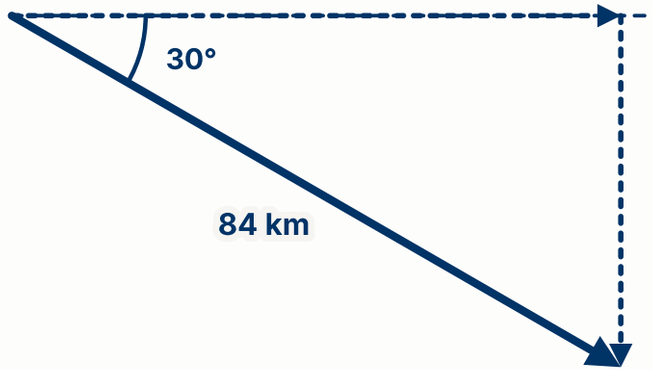
N-S Component:

$$\sin\theta = \frac{\text{opp}}{\text{hyp}}$$

$$\text{opp} = (\text{hyp})(\sin\theta)$$

$$\text{opp} = (84\ \text{km})(\sin 30^\circ)$$

$$\mathbf{opp = 42.0\ km}$$



E-W Component:

$$\cos\theta = \frac{\text{adj}}{\text{hyp}}$$

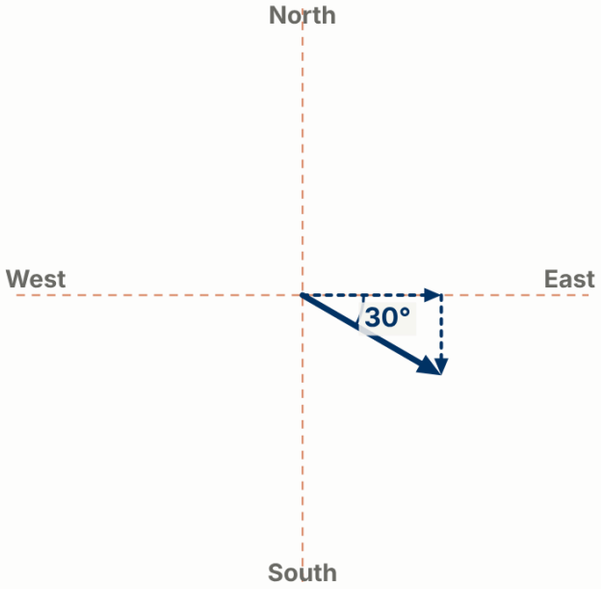
$$\text{adj} = (\text{hyp})(\cos\theta)$$

$$\text{adj} = (84\ \text{km})(\cos 30^\circ)$$

$$\mathbf{adj = 72.7\ km}$$

Vectors

Vector Resolution Method



84 km 30.0° south of east

1 Draw the vector being resolved
2 Draw the N-S component
3 Draw the E-W component
4 Move one component to form a right-angle triangle
5 Use trigonometry to calculate the components

$$\text{N-S: opp} = (\text{hyp})(\sin\theta) = (84\ \text{km})(\sin 30^\circ) = \mathbf{42.0\ km\ south}$$

$$\text{E-W: adj} = (\text{hyp})(\cos\theta) = (84\ \text{km})(\cos 30^\circ) = \mathbf{72.7\ km\ east}$$

with N and E positive: N-S −42.0 km, E-W 72.7 km

Magnitude **84 km**

Direction **-30°**

Step **5 of 5**

Direction is measured anticlockwise from east. The example, 84 km at 30° south of east, resolves into **42.0 km south** and **72.7 km east**.

Vectors

Vector Resolution Method

- so far, we have reviewed how to resolve a vector into its directional components. Now we will learn how to use this tool to add two or more vectors together (vector composition)
- the Vector Resolution Method of Vector Composition (adding of vectors) involves breaking down each vector being added into its directional components, adding all of the respective components, then combining the new components to arrive at the resultant

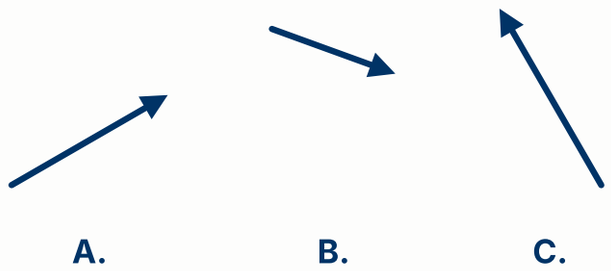
Vectors

Vector Resolution Method

for example

Calculate the sum of the following direction vectors:

- Vector **A** has a magnitude of 433 km directed 30° north of east
- Vector **B** has a magnitude of 315 km directed 20° south of east
- Vector **C** has a magnitude of 489 km directed 60° north of west



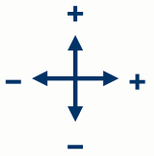
Vectors

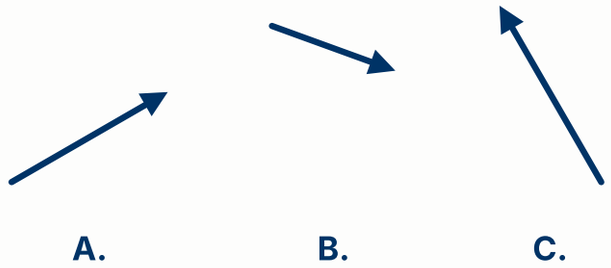
Vector Resolution Method

solution

- calculate the N-S and E-W components of each vector
- add the components, respectively
- calculate the new resultant

Vector	N-S	E-W
A.		
B.		
C.		
Sum		





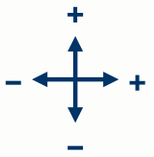
Vectors

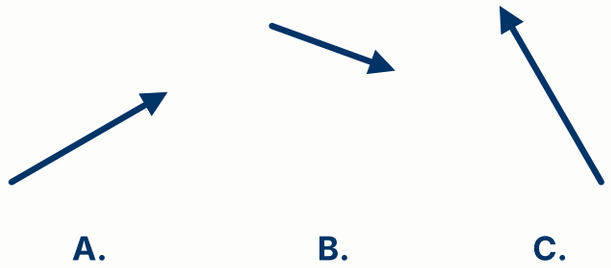
Vector Resolution Method

Group solution

- resolve vectors A, B and C into component vectors

Vector	N-S	E-W
A.		
B.		
C.		
Sum		





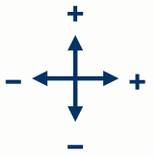
Vectors

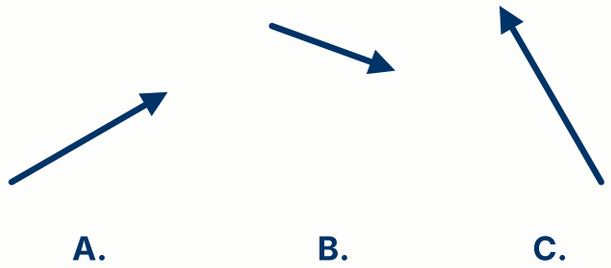
Vector Resolution Method

solution

- Draw the N-S and E-W components of vector A

Vector	N-S	E-W
A.		
B.		
C.		
Sum		





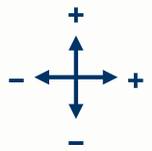
Vectors

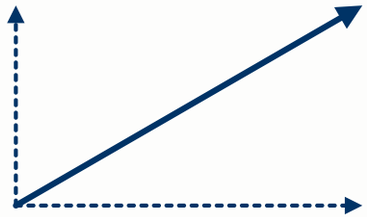
Vector Resolution Method

solution

- Draw the N-S and E-W components of vector A

Vector	N-S	E-W
A.		
B.		
C.		
Sum		





A.

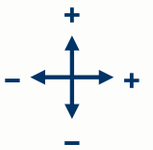
Vectors

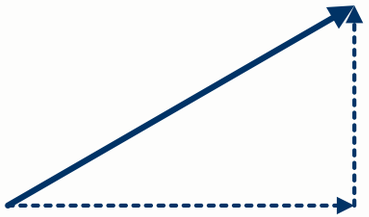
Vector Resolution Method

solution

- Imagine moving one of the component vectors to form a right-angle triangle.

Vector	N-S	E-W
A.		
B.		
C.		
Sum		





A.

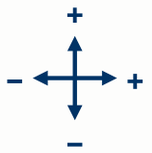
Vectors

Vector Resolution Method

solution

- Use the appropriate trigonometric functions to calculate the length of the components.

Vector	N-S	E-W
A.	**216.5 km**	**375.0 km**
B.		
C.		
Sum		



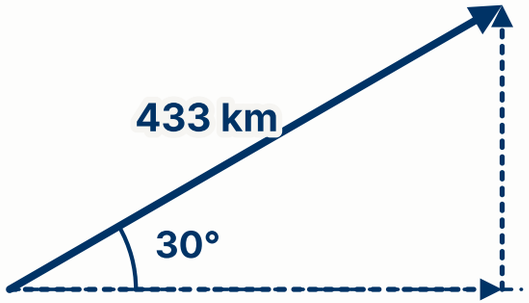
N-S Component:

$$\sin\theta = \frac{\text{opp}}{\text{hyp}}$$

$$\text{opp} = (\text{hyp})(\sin\theta)$$

$$\text{opp} = (433\ \text{km})(\sin 30^\circ)$$

$$\mathbf{opp = 216.5\ km\ North}$$



E-W Component:

$$\cos\theta = \frac{\text{adj}}{\text{hyp}}$$

$$\text{adj} = (\text{hyp})(\cos\theta)$$

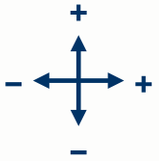
$$\text{adj} = (433\ \text{km})(\cos 30^\circ)$$

$$\mathbf{adj = 375.0\ km\ east}$$

Vectors

Vector Resolution Method

Vector	N-S Component	E-W Component
A.	216.5 km	375.0 km
B.		
C.		
Sum		



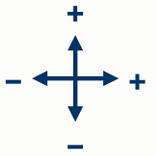
Vectors

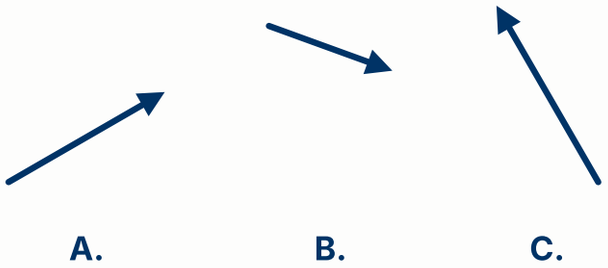
Vector Resolution Method

solution

- Perform the same steps to calculate the N-S and E-W components of Vector **B**.

Vector	N-S	E-W
A.	216.5 km	375.0 km
B.		
C.		
Sum		





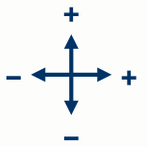
Vectors

Vector Resolution Method

solution

- Draw the N-S and E-W components of vector B

Vector	N-S	E-W
A.	216.5 km	375.0 km
B.		
C.		
Sum		





B.

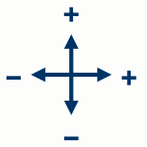
Vectors

Vector Resolution Method

solution

- Imagine moving one of the component vectors to form a right-angle triangle.

Vector	N-S	E-W
A.	216.5 km	375.0 km
B.		
C.		
Sum		





B.

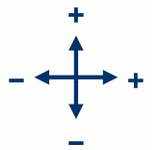
Vectors

Vector Resolution Method

solution

- Use the appropriate trigonometric functions to calculate the length of the components.

Vector	N-S	E-W
A.	216.5 km	375.0 km
B.	**−107.7 km**	**296.0 km**
C.		
Sum		



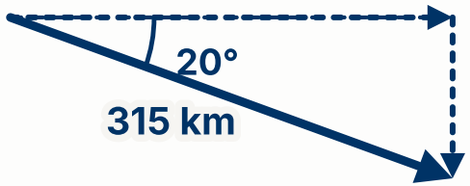
N-S Component:

$$\sin\theta = \frac{\text{opp}}{\text{hyp}}$$

$$\text{opp} = (\text{hyp})(\sin\theta)$$

$$\text{opp} = (315\ \text{km})(\sin 20^\circ)$$

$$\mathbf{opp = 107.7\ km\ south}$$



E-W Component:

$$\cos\theta = \frac{\text{adj}}{\text{hyp}}$$

$$\text{adj} = (\text{hyp})(\cos\theta)$$

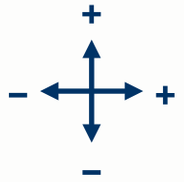
$$\text{adj} = (315\ \text{km})(\cos 20^\circ)$$

$$\mathbf{adj = 296.0\ km\ east}$$

Vectors

Vector Resolution Method

Vector	N-S Component	E-W Component
A.	216.5 km	375.0 km
B.	−107.7 km	296.0 km
C.		
Sum		



- the N-S component of Vector **B** is negative because we have defined N and E as positive and W and S as negative

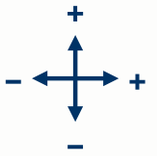
Vectors

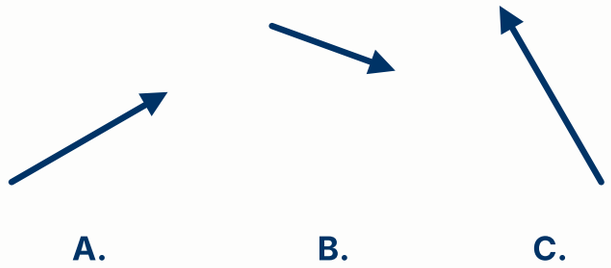
Vector Resolution Method

solution

- Perform the same steps one more time to calculate the vertical and horizontal components of vector **C**.

Vector	N-S	E-W
A.	216.5 km	375.0 km
B.	−107.7 km	296.0 km
C.		
Sum		





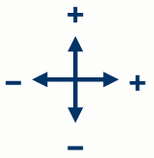
Vectors

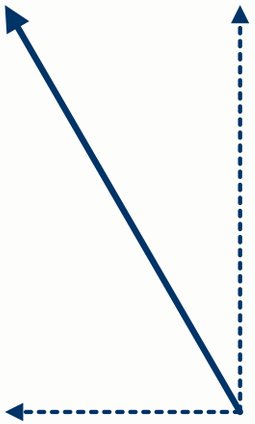
Vector Resolution Method

solution

- Draw the N-S and E-W components of vector C

Vector	N-S	E-W
A.	216.5 km	375.0 km
B.	−107.7 km	296.0 km
C.		
Sum		





C.

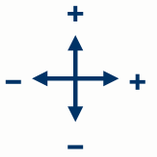
Vectors

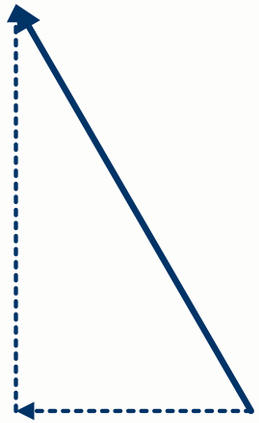
Vector Resolution Method

solution

- Imagine moving one of the component vectors to form a right-angle triangle.

Vector	N-S	E-W
A.	216.5 km	375.0 km
B.	−107.7 km	296.0 km
C.		
Sum		





C.

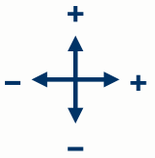
Vectors

Vector Resolution Method

solution

- Use the appropriate trigonometric functions to calculate the length of the components.

Vector	N-S	E-W
A.	216.5 km	375.0 km
B.	−107.7 km	296.0 km
C.	**423.5 km**	**−244.5 km**
Sum		



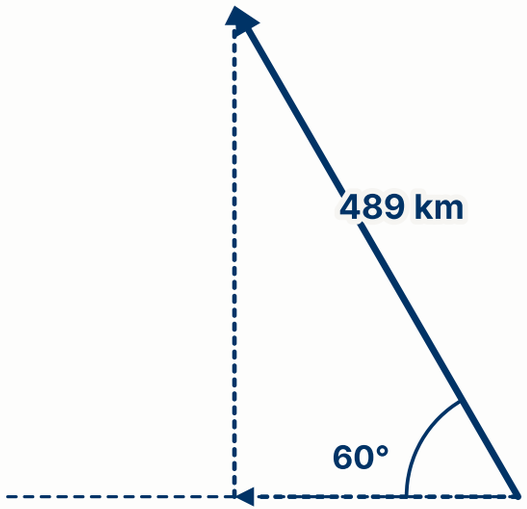
N-S Component:

$$\sin\theta = \frac{\text{opp}}{\text{hyp}}$$

$$\text{opp} = (\text{hyp})(\sin\theta)$$

$$\text{opp} = (489\ \text{km})(\sin 60^\circ)$$

$$\mathbf{opp = 423.5\ km\ North}$$



E-W Component:

$$\cos\theta = \frac{\text{adj}}{\text{hyp}}$$

$$\text{adj} = (\text{hyp})(\cos\theta)$$

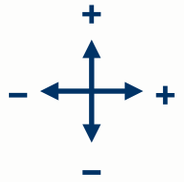
$$\text{adj} = (489\ \text{km})(\cos 60^\circ)$$

$$\mathbf{adj = 244.5\ km\ west}$$

Vectors

Vector Resolution Method

Vector	N-S Component	E-W Component
A.	216.5 km	375.0 km
B.	−107.7 km	296.0 km
C.	423.5 km	−244.5 km
Sum	**532.3 km**	**426.5 km**



- The E-W component of Vector **C** is negative because we have defined N and E as positive and W and S as negative

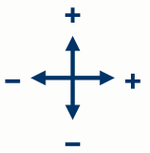
Vectors

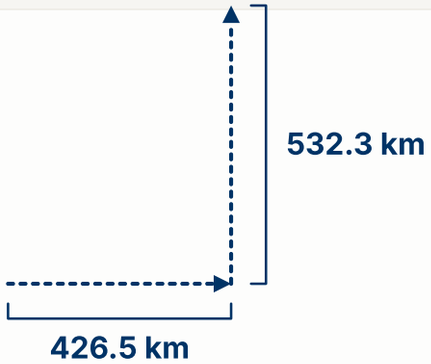
Vector Resolution Method

solution

- Combine the new N-S and E-W components. Imagine moving one of the component vectors to form a right-angle triangle.

Vector	N-S	E-W
A.	216.5 km	375.0 km
B.	−107.7 km	296.0 km
C.	423.5 km	−244.5 km
Sum	532.3 km	426.5 km





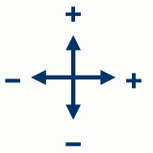
Vectors

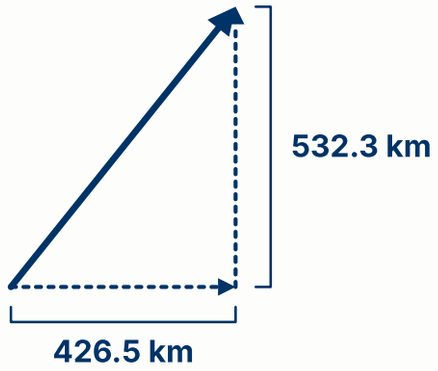
Vector Resolution Method

solution

- Determine the magnitude of the resultant using the Pythagorean Theorem.

Vector	N-S	E-W
A.	216.5 km	375.0 km
B.	−107.7 km	296.0 km
C.	423.5 km	−244.5 km
Sum	532.3 km	426.5 km





Magnitude of the Resultant:

$$A^2 + B^2 = C^2$$

$$(532.3\ \text{km})^2 + (426.5\ \text{km})^2 = C^2$$

$$\mathbf{C = 682.1\ km}$$

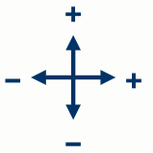
Vectors

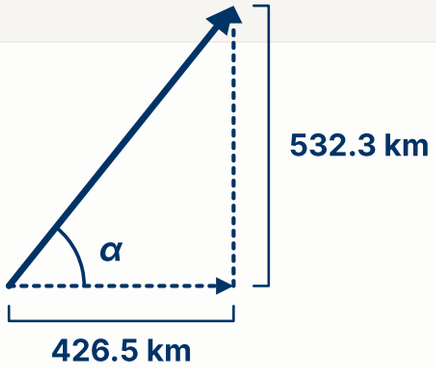
Vector Resolution Method

solution

- Determine the direction of the resultant using one of the trigonometric functions (any of them can be used because we know the length of all three sides of the triangle).

Vector	N-S	E-W
A.	216.5 km	375.0 km
B.	−107.7 km	296.0 km
C.	423.5 km	−244.5 km
Sum	532.3 km	426.5 km





Direction of the Resultant:

$$\tan\alpha = \frac{\text{opp}}{\text{adj}}$$

$$\tan\alpha = \frac{532.3\ \text{km}}{426.5\ \text{km}}$$

$$\tan\alpha = 1.25$$

$$\boldsymbol{\alpha = \tan^{-1}(1.25) = 51.3^{\circ}}$$

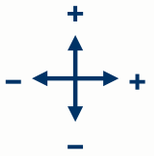
Vectors

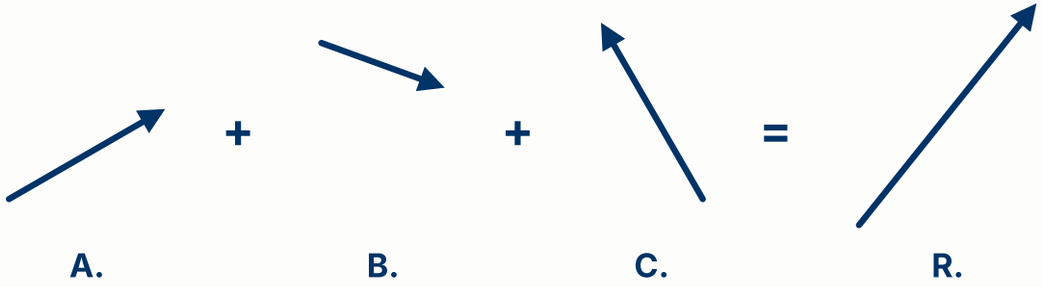
Vector Resolution Method

solution

- calculate the N-S and E-W components of each vector
- add the components, respectively
- calculate the new resultant

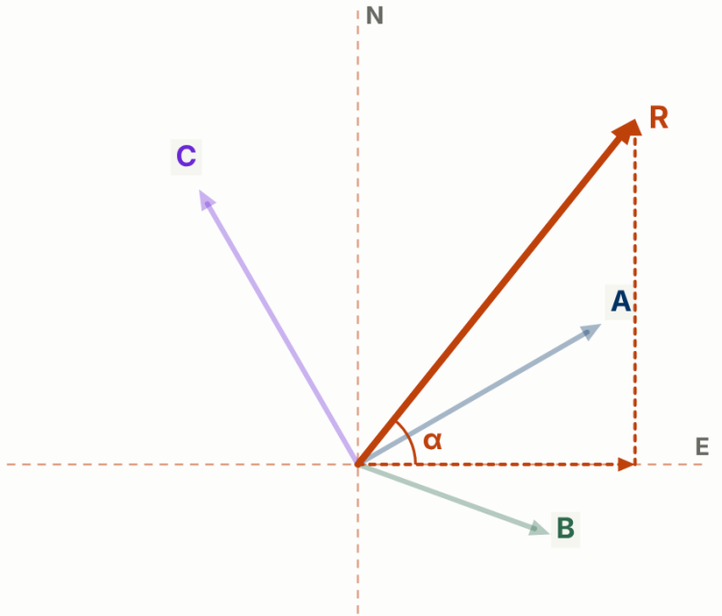
Vector	N-S	E-W
A.	216.5 km	375.0 km
B.	−107.7 km	296.0 km
C.	423.5 km	−244.5 km
Sum	532.3 km	426.5 km





Vectors

Vector Resolution Method



Vector	N-S Component	E-W Component
A.	216.5 km	375.0 km
B.	−107.7 km	296.0 km
C.	423.5 km	−244.5 km
Sum	532.3 km	426.5 km

6. combine the sums into the resultant

$C^2 = (532.3\ \text{km})^2 + (426.5\ \text{km})^2 \qquad \mathbf{C = 682.1\ km}$

$\tan\alpha = \dfrac{532.3}{426.5} = 1.25 \qquad \boldsymbol{\alpha = 51.3^\circ}$

Step 6 of 6

Resolve each vector, add the N-S column and the E-W column, then combine the two sums. The resultant is **682.1 km at 51.3° north of east**.

Spatial and Temporal

time (/taɪm/)

- a non-spatial continuum in which events occur in apparently irreversible succession from the past through the present to the future

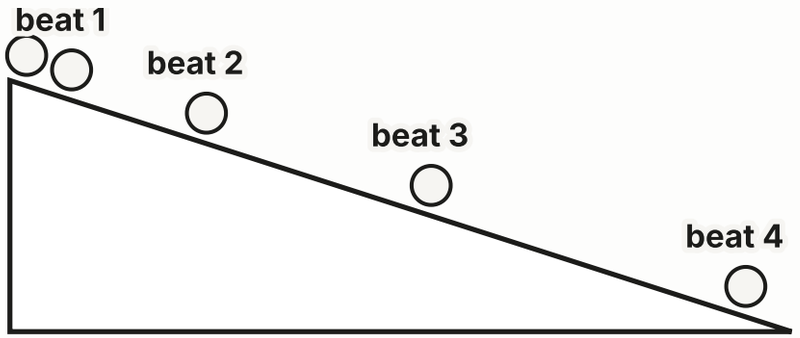
How was time measured before clocks existed?

Galileo Galilei (1564–1642)

We require an oscillating source that has a very consistent time between oscillations.

Modern day time keeping is possible through monitoring atomic decay and crystal oscillations.

Most computers use internal quartz crystals and synchronize to external atomic clocks





Galileo used his own pulse as a measure of time

Speed and Velocity

Linear Speed

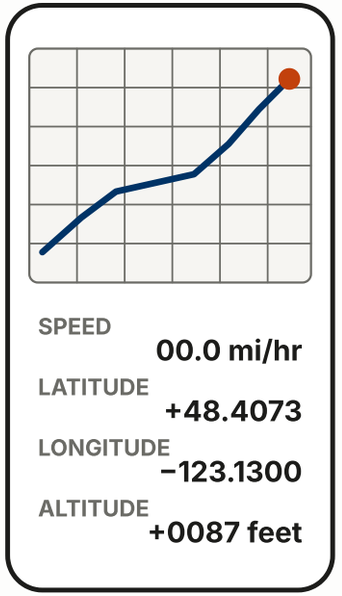
- represented by the letter *s*
- the distance covered divided by the time taken to cover it
- scalar variable (magnitude only)
- SI units: m/s

$$s = \frac{\ell}{\Delta t}$$

Linear Velocity

- represented by the letter *v*
- the displacement that occurs during a given period of time
- vector variable (magnitude and direction)
- SI units: m/s

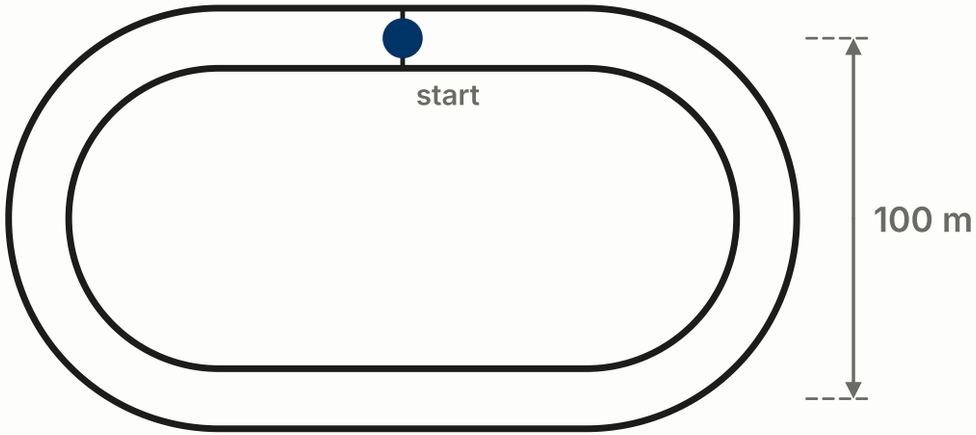
$$v = \frac{\Delta d}{\Delta t}$$



Speed and Velocity

Example Problem

A runner completes 1.5 laps of a 400 m track in 37.8 s. What are her speed and velocity?

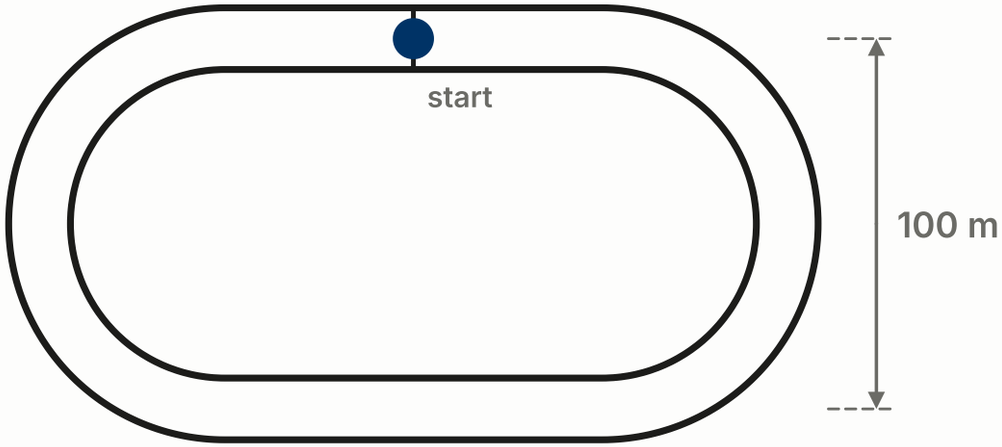


Speed and Velocity

Solution

Speed is distance divided by the change in time. Therefore, her speed is:

$$\frac{600\ \text{m}}{37.8\ \text{s}} = \mathbf{15.9\ m/s}$$

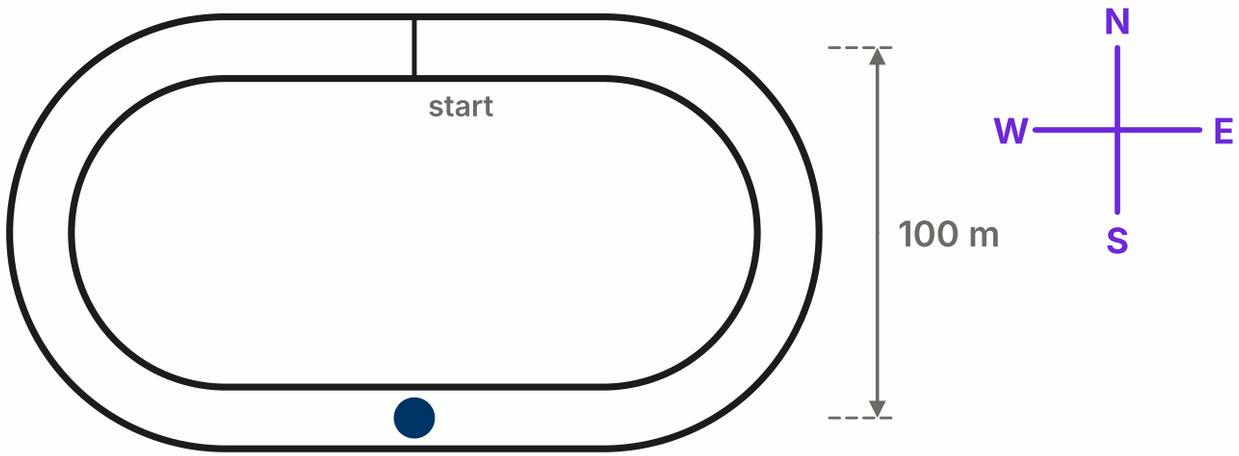


Speed and Velocity

Solution

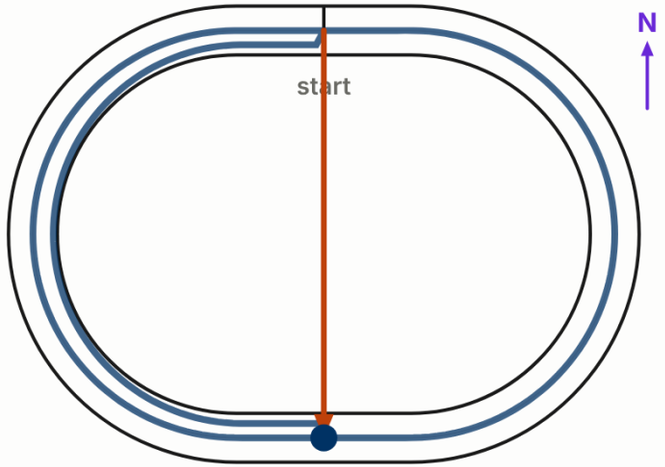
Velocity is displacement divided by the change in time. Therefore, her velocity is:

$$\frac{100\ \text{m}}{37.8\ \text{s}} = \mathbf{2.6\ m/s\ south}$$



Speed and Velocity

The same runner, with a clock



1.50 laps

time
37.8 s

distance run
600 m

displacement
100.0 m south

$\text{speed} = \text{distance}/\text{time}$
15.9 m/s

$\text{velocity} = \text{displacement}/\text{time}$
2.6 m/s south

Distance run — **600 m**

▶ Run

1.5 laps in 37.8 s. **Speed** is the distance over the time; **velocity** is the displacement over the time. At the finish: 15.9 m/s and 2.6 m/s south.

Acceleration

Linear Acceleration

- represented by the letter *a*
- the rate of change in velocity
- vector variable (magnitude & direction)
- SI units: m/s^2 (think of m/s per s)

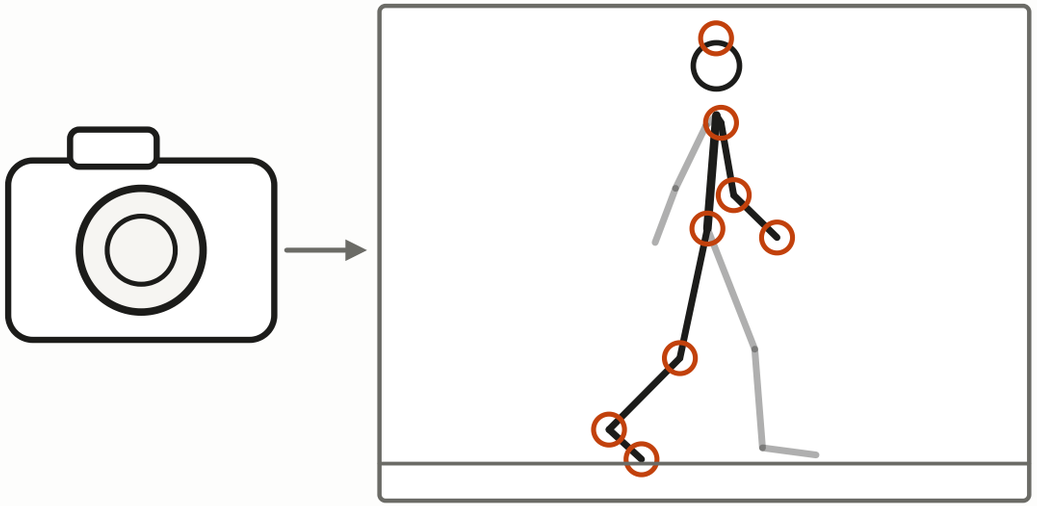
$$\boldsymbol{a} = \frac{\boldsymbol{\Delta v}}{\boldsymbol{\Delta t}}$$

$$= \frac{\text{change in velocity}}{\text{change in time}}$$

- for example, if a runner increases his speed by 2 m/s every second, his acceleration is 2 m/s per second or 2 m/s^2
- How does acceleration relate to sprinting performance?
- In the previous example we talked about the speed of an individual but we didn't know if they were faster at some points in the run and slower at other points.
- Now that we have acceleration we can determine when someone is speeding up, slowing down, starting or stopping

how do we measure basic linear kinematics

- Video analysis



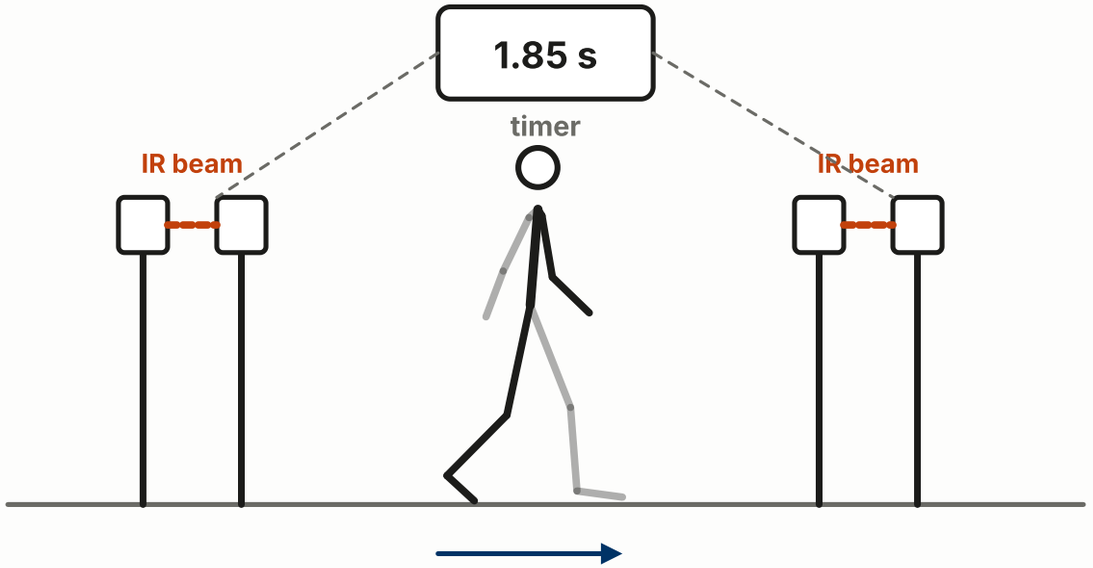
Source: walking stride, subject SUSU-30 — SUSU youth motion dataset · UVic · CC BY 4.0

Kinovea

how do we measure basic linear kinematics

Infrared timing gates

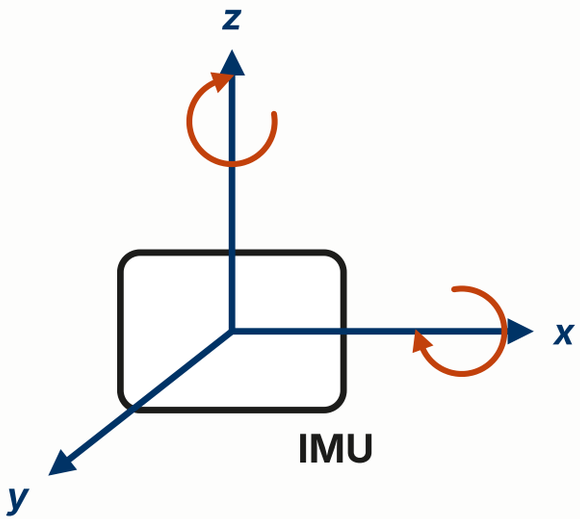
Photogates use the same technology that your TV remote uses. On one side of the small slot is an LED (like a small light bulb) that emits a beam of infrared light. On the other side is a detector that sees infrared light. You can’t see the light beam because human eyes cannot see infrared light. However, the electronic detector instantly senses when anything blocks the light beam. That is how the timer box is started and stopped



how do we measure basic linear kinematics

Inertial Measurement Units (IMU)

- Accelerometers
- Gyroscopes
- Magnetometers



accelerometers: along x, y, z

gyroscopes: about x, y, z

magnetometers: which way is north



100 m final, Beijing 2008

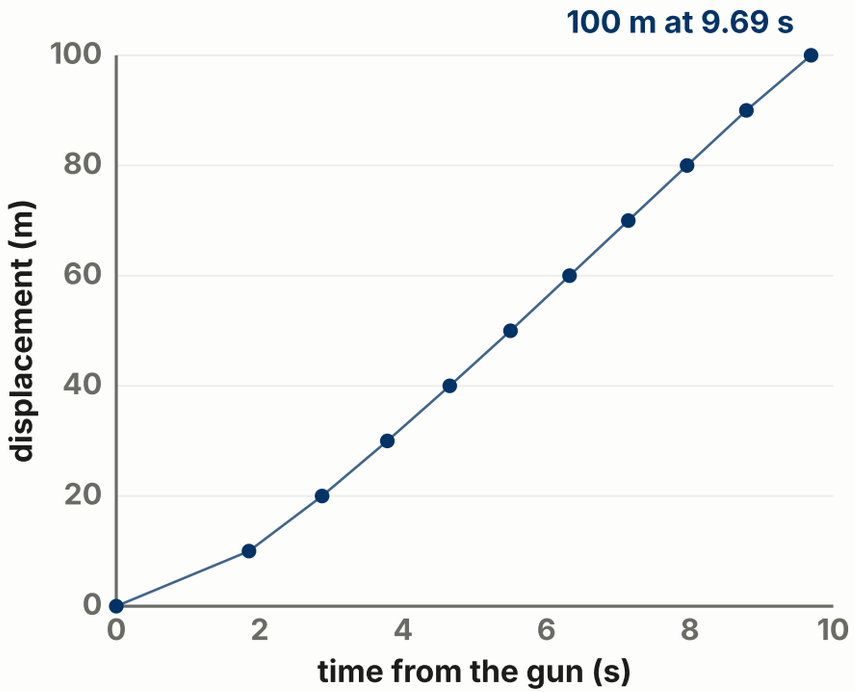
Usain Bolt · 9.69 s

Acceleration

Displacement: where he is, and when

Add the ten split times up from the gun and the race becomes eleven moments: where Bolt is, and when he is there.

distance (m)	time (s)
0	0.00
10	1.85
20	2.87
30	3.78
40	4.65
50	5.50
60	6.32
70	7.14
80	7.96
90	8.79
100	9.69



Source: split times: Usain Bolt, 100 m final, Beijing 2008

Acceleration

Velocity: how fast the displacement changes

Velocity is the change in displacement divided by the change in time — the **slope** of the displacement graph.

$$v = \frac{\Delta d}{\Delta t}$$

The whole race:

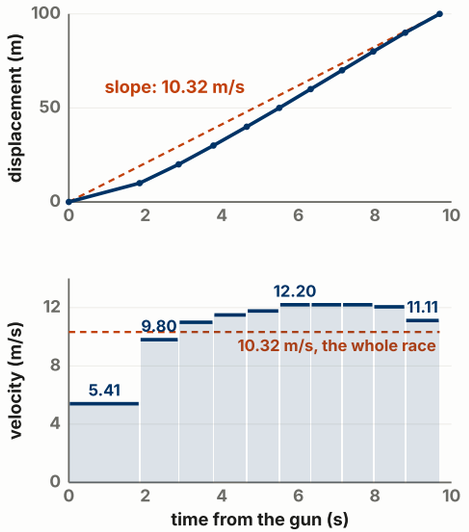
$$v = \frac{100\ \text{m} - 0\ \text{m}}{9.69\ \text{s} - 0\ \text{s}} = \mathbf{10.32\ m/s}$$

Each 10 m section:

$$v = \frac{10\ \text{m}}{1.85\ \text{s}} = \mathbf{5.41\ m/s}$$

$$v = \frac{10\ \text{m}}{1.02\ \text{s}} = \mathbf{9.80\ m/s}$$

One number for the whole race; ten for the sections, and they show him getting faster, holding his speed, and easing off at the line.



Source: split times: Usain Bolt, 100 m final, Beijing 2008

Acceleration

Example Problem

Given the kinematic data below, determine Usain Bolt’s average acceleration throughout the 100 m race.

Displacement	0 m		100 m
Time	0 s		9.69 s
Velocity	0 m/s		11.93 m/s

Source: split times: Usain Bolt, 100 m final, Beijing 2008

Acceleration

Solution:

$$a = \frac{\Delta v}{\Delta t}$$

$$a = \frac{11.93\ \mathrm{m/s} - 0\ \mathrm{m/s}}{9.69\ \mathrm{s} - 0\ \mathrm{s}}$$

$$\mathbf{a = 1.23\ m/s^2}$$

Throughout race, Bolt's average acceleration was 1.23 m/s^2.

Instantaneous vs. Average Values

- **What we have been calculating are averages over a certain period of time**
- **instantaneous** values can be thought of as **average** values that are calculated over infinitely small periods of time (i.e. as close as the time that you use to calculate the average $t_2 - t_1$ can get to being 0)
- an average value does not provide accurate information about rate of change
- for example, Bolt's velocity and acceleration were likely to have been changing within each 50 m section of the 100 m race, however, we only have one average value the whole race.

Acceleration

Example Problem

- So what if we had more data from the race?

Given the kinematic data below, determine Usain Bolt's average acceleration throughout the first and second halves of the 100 m race.

Displacement	0 m	50 m	100 m
Time	0 s	5.50 s	9.69 s
Velocity	0 m/s	9.09 m/s	11.93 m/s

Source: split times: Usain Bolt, 100 m final, Beijing 2008

Acceleration

Solution: the 1st 50 m

$$a = \frac{\Delta v}{\Delta t}$$

$$a = \frac{9.09\ \text{m/s} - 0\ \text{m/s}}{5.50\ \text{s} - 0\ \text{s}}$$

$$\mathbf{a = 1.65\ m/s^2}$$

Throughout the first 50 m, Bolt's average acceleration was 1.65 m/s^2.

Solution: the 2nd 50 m

$$a = \frac{\Delta v}{\Delta t}$$

$$a = \frac{11.93\ \text{m/s} - 9.09\ \text{m/s}}{9.69\ \text{s} - 5.50\ \text{s}}$$

$$\mathbf{a = 0.68\ m/s^2}$$

Throughout the second 50 m, Bolt's average acceleration was 0.68 m/s^2.

Acceleration

Example Problem

- So what if we had even more data?

Given the kinematic data, determine Usain Bolt's average acceleration throughout each section of the race.

Distance	Split Time	Velocity
0	0.17	
10	1.85	5.41
20	1.02	9.8
30	0.91	10.99
40	0.87	11.49
50	0.85	11.76
60	0.82	12.2
70	0.82	12.2
80	0.82	12.2
90	0.83	12.05
100	0.9	11.11
Total	9.69	10.32

Source: split times: Usain Bolt, 100 m final, Beijing 2008

Acceleration

Example Problem

The 1st split time at 0 distance is Usain's reaction time before he starts moving.

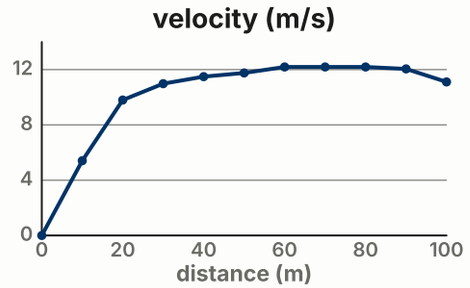
Each split time is the time between the previous and current distance

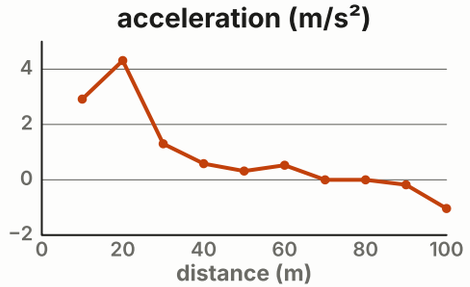
Distance	Split Time	Velocity
0	0.17	
10	1.85	5.41
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50	0.85	11.76
60	0.82	12.2
70	0.82	12.2
80	0.82	12.2
90	0.83	12.05
100	0.9	11.11
Total	9.69	10.32

Source: split times: Usain Bolt, 100 m final, Beijing 2008

Acceleration

meters	split time	velocity	acceleration
0	0.17		
10	1.85	5.41	2.92
20	1.02	9.80	4.31
30	0.91	10.99	1.30
40	0.87	11.49	0.58
50	0.85	11.76	0.32
60	0.82	12.20	0.52
70	0.82	12.20	0.00
80	0.82	12.20	0.00
90	0.83	12.05	−0.18
100	0.90	11.11	−1.04
Total	9.69		

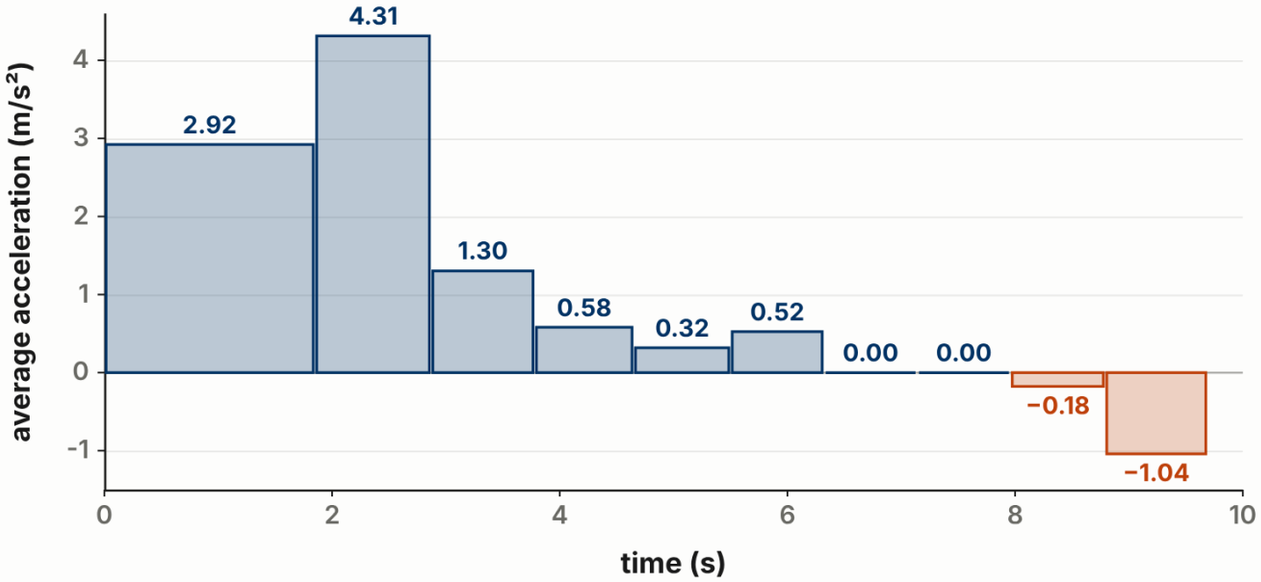




Source: split times: Usain Bolt, 100 m final, Beijing 2008

Acceleration

Average acceleration, over shorter and shorter sections



each 10 m section

0–10 m	**2.92 m/s²**
10–20 m	**4.31 m/s²**
20–30 m	**1.30 m/s²**
30–40 m	**0.58 m/s²**
40–50 m	**0.32 m/s²**
50–60 m	**0.52 m/s²**
60–70 m	**0.00 m/s²**
70–80 m	**0.00 m/s²**
80–90 m	**−0.18 m/s²**
90–100 m	**−1.04 m/s²**

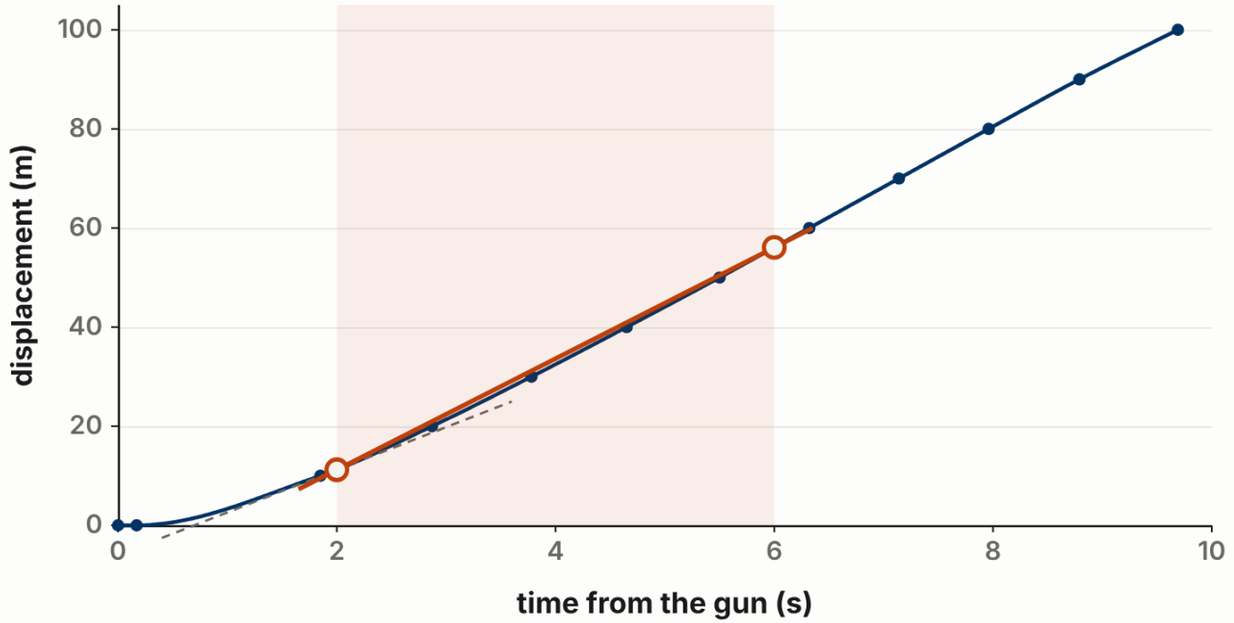
The whole race | Two halves | **Every 10 m**

One average says 1.23 m/s². Two say he accelerated harder in the first half. Ten show **where**: hardest in the first 20 m, close to zero from 60 to 80 m, and negative at the finish.

Source: split times: Usain Bolt, 100 m final, Beijing 2008

Acceleration

Shorter and shorter: the slope at an instant



the section
2.00 s to 6.00 s
Δt = 4.00 s

average velocity over it
Δd / Δt = 44.87 m / 4.00 s
11.22 m/s

the tangent at 2.00 s
8.60 m/s
off by 2.61 m/s

Moment t_o **2.00 s** Section Δt **4.00 s**

▶ Shrink it

Over 4.00 s from 2.00 s the average velocity is **11.22 m/s**; the slope of the tangent there is **8.60 m/s**. Shrink the section and the first number settles on the second: the velocity at an instant is the limit of the average as Δt goes to 0. Do the same to the velocity and you get the acceleration. That is the calculus and kinematics lecture.

Source: split times: Usain Bolt, 100 m final, Beijing 2008 · smooth curve fitted through them

Sources

Where the measurements on these slides came from

> Every figure in this deck that shows a measurement or a model says underneath it where that came from. The full entries are here.

SUSU youth motion dataset	SUSU youth motion dataset, public release v2. University of Victoria. Zenodo, doi:10.5281/zenodo.20504820 (all versions: 10.5281/zenodo.20504819). Licensed CC BY 4.0. Data descriptor in preparation. https://doi.org/10.5281/zenodo.20504820
character rig	Character mesh and skeleton from Mixamo (Adobe Inc.), used under the Mixamo licence. Every pose on these slides was set from joint angles here and rendered here; no Mixamo animation is reproduced. https://www.mixamo.com
split times	Usain Bolt, men’s 100 m final, Games of the XXIX Olympiad, Beijing, 16 August 2008: reaction time and 10 m split times as given on the lecture slides. Velocities and accelerations on these slides are computed from those splits.